

Unit 3 Overview: Understanding Rational Numbers

Unit Narrative



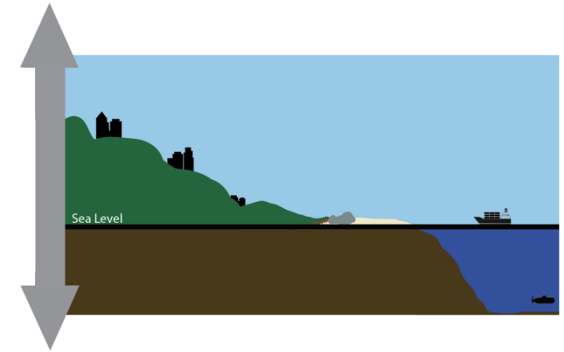
In this unit, students are introduced to negative rational numbers. They learn how to plot, order, compare, and find opposites of rational numbers. This is students' first formal introduction to negative numbers, extending the number line and their knowledge of performing operations and comparing positive values, and interpreting the meaning of rational numbers in various contexts. Students are also introduced to absolute value for the first time, including interpreting its meaning in various contexts and using absolute values to solve problems.

In Section 1, students learn how to plot, order, and compare rational numbers. Students develop an understanding of the sets of numbers (rational, irrational, integers, whole, and natural numbers) in the real-number system. They also plot rational numbers on horizontal and vertical number lines. Next, students explore and compare numbers and their opposites. Students also learn to interpret the meaning of zero in various real-world contexts. Knowledge of equivalent representations of rational numbers from Grade 6 Accelerated Unit 2 are relied on as they interpret and rewrite rational numbers in their equivalent forms (fraction, decimal, percent) to make comparisons and find locations on the number line.

Section 2 focuses on real-world contexts, where students interpret positive and negative values in real-world contexts. Next, they compare quantities and their opposites on horizontal or vertical lines, and explain the meaning of "zero" within the context. Subsequently, students use the structure of a horizontal and vertical number line to plot rational numbers representing real-world contexts.

In Section 3, students are introduced to the meaning of absolute value. They find the absolute values of rational numbers and learn to interpret their meaning in various contexts. Next, students compare and perform operations on the absolute values of two numbers. They apply their knowledge and skills to interpret and perform operations with absolute values in real-world contexts.

This unit lays the foundation for Grade 6 Accelerated Unit 11 where students will perform operations with rational numbers and solve real-world problems with rational numbers.



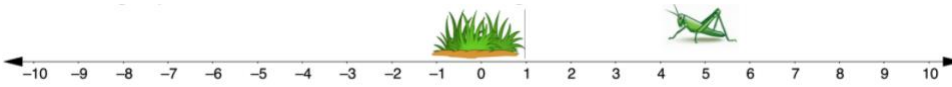
Unit At-A-Glance

Section 1: Rational Numbers and the Number Line	Benchmarks Addressed	Lesson 1	Rational Numbers and the Number Line – Part 1
	MA.6.NSO.1.1 MA.6.NSO.1.2	Lesson 2	Rational Numbers and the Number Line – Part 2
		Lesson 3	Plotting, Ordering, and Comparing Rational Numbers – Part 1
		Lesson 4	Plotting, Ordering, and Comparing Rational Numbers – Part 2
Section 2: Rational Numbers in Context	Benchmarks Addressed	Lesson 5	Rational Numbers in Real-World Context
	MA.6.NSO.1.2	Lesson 6	Comparing a Quantity and Its Opposite in Real-World Context
Section 3: Absolute Value	Benchmarks Addressed	Lesson 7	Absolute Value
	MA.6.NSO.1.3 MA.6.NSO.1.4	Lesson 8	Comparing and Performing Operations with Absolute Value
		Lesson 9	Operations with Absolute Value in the Real World



Differentiation

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Engagement <i>The “WHY” of learning</i>	- The real-world problems throughout the unit (for example, the jumping bugs Exploration in Lesson 7) help students gain an appreciation for the utility of the mathematics they are learning. The contexts presented will also engage students if they are related to their interests. If needed, you can modify the contexts in the problems based on the interests of your students. - In the components where students are asked to work independently, consider allowing students who struggle to pair up with a partner. This will ensure students are engaged in productive struggle and sustain effort. - Several Wrap-Up components ask students to self-regulate by reflecting on their mastery of the learning goals and develop a plan to fill in any gaps in their learning. 
Representation <i>The “WHAT” of learning</i>	- Allow students to represent their work using vertical or horizontal number lines throughout this unit. The number lines provide scaffolding to build conceptual understanding of plotting, comparing, and ordering rational numbers. - The Activity in Lesson 2 provides visual and kinesthetic entry points as students create foldable number lines on paper. - Auditory entry points are available to students in the Collaboration activities throughout the unit. Partner discussion will help students gain confidence with and understanding of the learning targets.
Action & Expression <i>The “HOW” of learning</i>	- Several components include constructive-response problems that require students to explain why or how they solved the problem. The teacher guides include alternative methods for students who struggle expressing themselves in writing, such as using an audio recorder or speech-to-text software. - In the Wrap-Ups throughout the unit, allow students to demonstrate their mastery of the learning targets using different methods, such as writing a reflective self-assessment, solving example problems, or drawing a visual model. - I can compare the absolute value of two numbers. I need help. I'm getting there. I understand and have evidence. I can't get started. <i>Answers will vary.</i> - I can perform operations that involve the absolute value of two numbers. I need help. I'm getting there. I understand and have evidence. I can't get started. <i>Answers will vary.</i>

ELL Information

Students solve real-world problems in many of the lessons in this unit. These problems can be text and reading heavy. When translating words into numeric expressions, encourage students to underline key words. Jose's video will be helpful for ELL students to watch for this topic because he will model these strategies. If ELL students respond well, consider recommending that Jose be their go-to Study Expert!

Because this unit has some text-heavy problems, “Mark Up Text” is a good activity to help ELL students begin feeling comfortable translating words into mathematical expressions.

- The teacher will provide students with the following symbols:
 - ✓ - background knowledge: they know what to do.
 - ? - confusing information: Is this needed to solve the problem?
 - ! - marks where a variable is needed to represent a number.
 - + - extra info in word problem: not needed.
- Using a small piece of text, the teacher can model for students how they could mark up before releasing students to do the same.
- Students will read the text slowly and carefully, focusing on one sentence at a time and marking text based on the key. Then they will turn to a partner to discuss the context and compare where they made marks.
- For the first few examples, have groups share their decisions out to the class. Then, guide students through turning the marked-up text into an expression. ELL students may find this a helpful intermediary step, so encourage them to use this strategy even as you release them to work independently.

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students.

- | | | |
|------------------|-------------------|----------------|
| • Absolute Value | • Natural number | • Real numbers |
| • Integer | • Rational number | • Whole number |

Additional Differentiated Support

Scaffolding Resources	On-Ramp
Human Number Lines and Real-World Number Lines Add hands-on learning opportunities using life-size number lines for components like Lesson 1 Activity 4 in this unit. Students can choose a number special to them between 0 and 20. Students should choose their number and determine whether they feel “negative” or “positive” to determine which side of the number line they will stand. Students are then challenged to make decisions for how “far away” they should stand from each other to represent the mental work of scaling the number line. This kinesthetic learning experience can be a fun and engaging way to begin developing students’ number senses using a number line.	On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills: - Number Lines: Fractions - Representing Fractions on a Number Line - Compare and Order Decimals with Models - Compare and Order Decimals with $>$, $=$, or $<$ - Comparing Fractions Using Symbols $>$, $=$, or $<$ - Comparing Fractions with Like Numerators and Denominators - Compare and Order Whole Numbers with $>$, $=$, or $<$



Section Overview

Section 1: Rational Numbers and the Number Line (Lessons 1 – 4)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order and compare rational numbers. <i>Clarification 1: Within this benchmark, the expectation is to plot, order and compare positive and negative rational numbers when given in the same form and to plot, order and compare positive rational numbers when given in different forms (fraction, decimal, percentage).</i> <i>Clarification 2: Within this benchmark, the expectation is to use symbols ($<$, $>$, or $=$).</i> MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context. <i>Clarification 1: Instruction includes vertical and horizontal number lines, context referring to distances, temperatures and finances and using informal verbal comparisons, such as, lower, warmer or more in debt.</i> <i>Clarification 2: Within this benchmark, the expectation is to compare positive and negative rational numbers when given in the same form.</i>	Lesson 1	- I can demonstrate an understanding of negative numbers. - I can recognize that a number in the real-number system can have a variety of classifications. - I can plot rational numbers on a horizontal and vertical number line.
	Lesson 2	- I can determine how far a number and its opposite are from zero. - I can represent a number and its opposite on a number line. - I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context.
	Lesson 3	- I can plot and compare rational numbers represented as a decimal or percent. - I can plot, compare, and order fractional representations of rational numbers.
	Lesson 4	I can plot, compare, and order rational numbers.

Section 2: Rational Numbers in Context (Lessons 5 – 6)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context. <i>Clarification 1: Instruction includes vertical and horizontal number lines, context referring to distances, temperatures and finances and using informal verbal comparisons, such as, lower, warmer or more in debt.</i> <i>Clarification 2: Within this benchmark, the expectation is to compare positive and negative rational numbers when given in the same form.</i>	Lesson 5	- I can determine positive and negative values in relation to real-world contexts. - I can plot negative rational values given a real-world context.
	Lesson 6	- I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context. - I can use the structure of a horizontal and vertical number line to plot rational numbers from a real-world context.

Section 3: Absolute Value (Lessons 7 – 9)

Benchmarks Addressed	Learning Targets	
MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers. • <i>Clarification 1: Instruction includes the connection of absolute value to mirror images about zero and to opposites.</i> • <i>Clarification 2: Instruction includes vertical and horizontal number lines and context referring to distances, temperature and finances.</i> MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value. • <i>Clarification 1: Absolute value situations include distances, temperatures and finances.</i> • <i>Clarification 2: Problems involving calculations with absolute value are limited to two or fewer operations.</i> • <i>Clarification 3: Within this benchmark, the expectation is to use integers only.</i>	Lesson 7	• I can find the absolute value of a rational number. • I can find the absolute value of a value and interpret it within the context.
	Lesson 8	• I can compare the absolute value of two numbers. • I can perform operations that involve the absolute values of two numbers.
	Lesson 9	• I can perform operations that involve the absolute value of two numbers.



Vertical Alignment

Grade 6 Accelerated Unit 3
Understanding Rational
Numbers



Grade 7 Accelerated Unit 2
Rational and Irrational Numbers

Prior Course(s)	Course Level	Future Course(s)
MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.	MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers. MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context. MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers. MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.	MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line. MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.

3.1 Rational Numbers and the Number Line – Part 1

Lesson Introduction

 Benchmark	MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.		 Learning Targets	- I can demonstrate an understanding of negative numbers. - I can recognize that a number in the real-number system can have a variety of classifications. - I can plot rational numbers on a horizontal and vertical number line.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How could you use a number line to represent positive and negative values relative to 0? - How does the direction, or how far left or how far right of 0, relate to the value of a number on a number line? - How can you scale number lines with tick marks to include fractional values?			
 Lesson Narrative	Students develop an understanding of negative numbers and their placement on the number line. Students explore the components of the real-number system. Students then plot rational numbers on the number line.			
 Component Summary	3.1.1 Warm-Up (~5 min.)	Create an equation equal to zero using positive whole numbers (<i>SE p. 75</i>)	3.1.5 Bring It Together (10 min.)	Guided practice plotting rational numbers on a number line (<i>SE p. 79</i>)
	3.1.2 Exploration (5 min.)	Explore the existence of negative numbers (<i>SE p. 76</i>)	3.1.6 Try It (5-10 min.)	Independent practice plotting rational numbers on horizontal and vertical number lines (<i>SE p. 80</i>)
	3.1.3 Guided Instruction (5-10 min.)	Learn about the real-number system (<i>SE p. 76</i>)	3.1.7 Your Turn! (5 min.)	Additional practice plotting rational numbers on horizontal and vertical number lines (<i>SE p. 81</i>)
	3.1.4 Activity (5-10 min.)	Extend a horizontal number line to include negative values (<i>SE p. 78</i>)	3.1.8 Wrap-Up (~5 min.)	Demonstrate understanding of negative numbers (<i>SE p. 81</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Natural numbers - Whole numbers - Integers - Rational numbers - Real numbers <u>Additional Terms:</u> N/A		(Optional) Different types of paper (patty, lined, graph, chart)	
			Additional Preparation	
			3.1.2 Exploration contains a video to accompany the question prompts.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may interpret larger negative numbers as having a larger value. - Students may believe negative numbers cannot be rational (because they are negative). - Students may have difficulty making the tick marks for the $\frac{1}{4}$ intervals.	- Explain and demonstrate that the negative numbers are a mirror image of the positive numbers on the number line to help clarify student thinking. 3.1.5 Bring It Together is intended to be guided after students have had an opportunity to explore the number line in the 3.1.4 Activity. Consider the language used when modeling how to plot values on the number line, particularly when plotting negative values, which is new for students. Use a think aloud to model the thinking involved so students can apply similar reasoning when practicing on their own in the components that follow. - If time is an issue, consider assigning 3.1.7 Try It! As homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 3.1.2 Exploration provides an opportunity to practice the Stronger and Clearer Each Time routine. After developing individual responses, students can work in small groups to critique each other’s ideas about negative numbers. Critiques should include suggestions for improving the clarity of each other’s verbal and written explanations.	MA.K12.MTR.4.1: Discussion and Reflection Students will work to effectively communicate mathematical ideas and vocabulary regarding negative numbers. In the 3.1.3 Guided Instruction, students can share their responses and discuss ideas and thoughts about the real-number system. Students should be encouraged to provide justification for responses.
 Differentiate Instruction	Math is what connects us all. Consider adding variety to 3.1.4 Activity with a human number line. Students can choose a number special to them between 0 and 20. Students should choose their number and determine whether they feel “negative” or “positive” to determine which side of the number line they will stand. Students are then challenged to make decisions for how “far away” they should stand from each other to represent the mental work of scaling the number line. This kinesthetic learning experience can be a fun and engaging way to begin developing students’ number senses using a number line. Note that incorporating this activity will require additional time. Consider when you would want to incorporate it and how you might adjust instruction to accommodate the additional time.	
Lesson Notes:		

3.1.1 Warm-Up: Creating Zero

Component Overview: Students use whole numbers one to nine to create an equation equal to zero. Students then switch equations with a partner. Partners check each other's equations and discuss their strategies.



3.1.1 Warm-Up: Creating Zero

1. Use only digits 1 to 9, at most one time each, so that the equation is true.

$$(\square + \square - \square + \square) - (\square + \square - \square + \square) = 0$$

Answers will vary.

Possible solution: $(9 + 1 - 8 + 5) - (7 + 4 - 6 + 2) = 0$

2. Switch your answer with a partner to check. Ask your partner their strategy for finding their answer and write their strategy below.

Answers will vary.



Require students to provide reasoning for methods used to solve the equation. Encourage students to clarify and critique during discussions. Students may change responses based on partner discussions.

3.1.2 Exploration: How Can Negative Numbers Exist?

Component Overview: Students watch “Negative Numbers: An Overview.” Students explore graphing negative numbers on a number line and real-world examples of negative numbers.



3.1.2 Exploration: How Can Negative Numbers Exist?

Negative numbers are numbers less than zero.
Examples include:

- a temperature below 0 degrees
 - the amount of money owed, or debt
 - a contestant’s score on Jeopardy (when someone misses a question, he/she loses points, so his/her score could be a negative amount)
1. Discuss the examples of negative numbers with your partner.
 2. Discuss with your partner which of the contexts in the table could have a value less than 0, then write an example or explain your thinking.

Context	Example or Explanation
Direction	<i>Answers will vary. An example would be if you are driving somewhere, but you drove in the wrong direction.</i>
Points in a game	<i>Answers will vary. An example is losing points in a game.</i>
Time	<i>Answers will vary. I don’t think it’s possible to have time less than 0.</i>



Playing the two-minute video “Negative Numbers: An Overview” prior to starting the activity provides visual and auditory entry points. Pause the video to explain that Mount Everest is so cold because it is the highest mountain above sea level. Students may also benefit from comparing and contrasting what they know about the weather in Alaska, Idaho, and Florida.



Stronger and Clearer Each Time
Consider requiring students to develop written responses prior to partner and/or group discussions. Then, have students provide each other with feedback. Next, students can revise their responses. Repeat this process as time allows for increasingly precise answers.

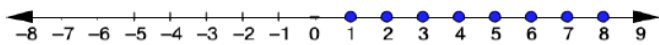
3.1.3 Guided Instruction: The Real Number System

Component Overview: Students learn about the sets of numbers in the real-number system. A visual entry point is provided as students observe where the natural numbers, whole numbers, and integers lie on the number line. Then, students explore examples of natural numbers, integers, and rational numbers.



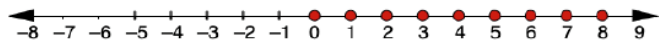
3.1.3 Guided Instruction: The Real Number System

The values in blue on the number line are some of the numbers in the set of **natural numbers**.



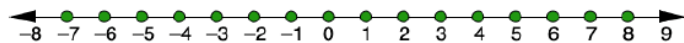
Natural numbers are the counting numbers
{1, 2, 3, 4, 5 ...}.

The values in red on the number line are some of the numbers in the set of **whole numbers**.



Whole numbers are natural numbers and zero.
{0, 1, 2, 3, 4, 5 ...}

The values in green on the number line are some of the values in the set of **integers**.



Integers are whole numbers and their opposites.
{... -4, -3, -2, -1, 0, 1, 2, 3, 4 ...}



What do you notice about the numbers as you move to the left of zero? How does the value of numbers compare as you move to the left of zero versus as you move to the right of zero?

1. Use the number lines to help to answer each of the following questions.
 - a. What do you notice about the set of natural numbers and the set of whole numbers?
Answers will vary. Only positive numbers are included in natural numbers. The natural numbers are whole numbers except for 0.

3.1.3 Guided Instruction: The Real Number System (continued)

- b. What do you notice about the set of integers?
Answers will vary. Positive and negative numbers are included.
- c. What type of number is -10 ?
Integer
- d. What type of number is 27 ?
Natural Number, Whole Number, and Integer.
- e. Describe the numbers that appear in all three sets.
Answers will vary. Natural numbers, or positive whole numbers, appear in all three sets.

Camaria wondered to herself, "What about numbers such as $3\frac{1}{2}$ and -1.75 ?"

These numbers are examples of **rational numbers**.



Rational numbers are numbers that can be expressed as the ratio of two integers.

2. Use the definition of rational numbers to write each as a ratio of two integers.

Num ber	Ratio of Two Integers
-1.5	$-\frac{3}{2}$
9	$\frac{9}{1}$

In your study of mathematics, you will learn about **irrational numbers**, which are numbers that cannot be written as a ratio of two integers.



The set of **real numbers** is the set of all rational and irrational numbers.



Ask students to share the ratios they wrote. Look for responses that have typical misconceptions. Some examples could be $\frac{-1.5}{1}$ or $\frac{9}{9}$. Also, consider sharing ratios that students may not have written such as $\frac{9}{-4}$ or $\frac{18}{2}$.

3.1.4 Activity: Horizontal Number Line

Component Overview: Students plot rational numbers on a number line. This activity provides a visual entry point as students develop a conceptual understanding of negative numbers.



3.1.4 Activity: Horizontal Number Line

Using a sheet of paper, follow the steps to make your own number line.

- Step 1: Use a straightedge or a ruler to draw a horizontal line. Mark the middle point of the line and label it 0.
- Step 2: To the right of 0, draw tick marks that are 1 centimeter apart. Label the tick marks 1, 2, 3...10.
- Step 3: Fold your paper so that a vertical crease goes through 0 and the two sides of the number line match up perfectly.
- Step 4: Use the fold to help you trace the tick marks that you already drew onto the opposite side of the number line. Unfold and label the tick marks $-1, -2, -3 \dots -10$. This represents the negative side of your number line.

Use the number line to answer each of the following questions.

- Describe how far away the number 5 is from 0.

*The number 5 is **five** units to the **right** of 0 on the number line.*

- Write a positive number and a negative number where each is farther away from 0 than the number 5.

Answers will vary.

Example: -6 and 6

- Describe how far away the number -2 is from 0.

*The number -2 is **two** units to the **left** of 0 on the number line.*

- Write a positive number and a negative number that are each farther away from 0 than -2 .

Answers will vary.

Example: -3 and 3



Visit the “Differentiate Instruction” note in Additional Planning Support for the Human Number Line activity. If using this activity, consider how you will adjust instruction to accommodate the additional time it would require.



Different types of paper can be used to answer practice problems. Lined, chart, and graph paper simplify making a number line, because they already include tick marks. Patty paper simplifies making a number line, because the positive side of the number line can be folded and rubbed onto the negative side of the number line.

3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

Component Overview: Students learn to plot positive and negative numbers on a number line. Then, students extend their knowledge and skills to include plotting fractions and decimals on the number line. Teacher should guide students through this component, modeling how to think through the relationships between the values to determine each value's location.

3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines

When plotting numbers on a number line, the sign of the number indicates which direction from 0 to move on the number line.

- On a horizontal number line, positive numbers are to the

○ right

○ left

 of zero.
- On a vertical number line, positive numbers are

○ above

○ below

 zero.
- On a horizontal number line, negative numbers are to the

○ right

○ left

 of zero.
- On a vertical number line, negative numbers are

○ above

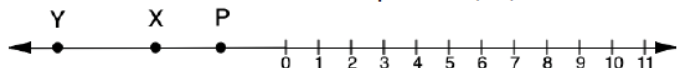
○ below

 zero.



Students can use the folded number line they created during the previous component as an additional scaffold when trying to understand how to locate each value on the number line, particularly when plotting negative values.

1. A number line is shown with points P , X , and Y .

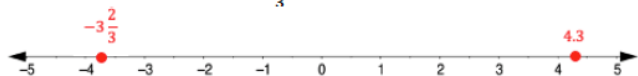


- What type of numbers are P , X , and Y ?
Negative numbers
- Estimate the location of points P , X , and Y .
-2, -4, -7

3.1.5 Bring It Together: Plotting Rational Numbers on Number Lines (continued)

2.

- a. Plot the values $-3\frac{2}{3}$ and 4.3 on the number line.



- b. How did the tick marks between each integer help you determine where to plot 4.3?

Answers will vary. The tick marks show integers and the halfway point between them. I know 4.3 is between 4 and 4.5, but closer to 4.5, so that was helpful to plot 4.3.

- c. How did the tick marks between each integer help you determine where to plot $-3\frac{2}{3}$?

Answers will vary. I know $-3\frac{2}{3}$ is between -3 and -4 on the number line, but closer to -4 . $-3\frac{2}{3}$ is also farther to left of 0 than -3.5 , so I used that to help me estimate its location.



How can making a tick mark that is halfway between each value on the number line help you plot fractions and decimals? How can you convert between fractions and decimals to assist in comparing their values?



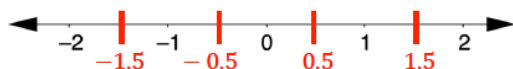
Think Aloud

For question 2, the teacher should think aloud as they plot the values on the number line to model the type of thinking students can use when making these decisions on their own in the next component.

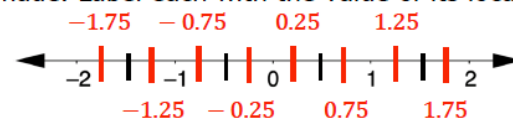
Based on student understanding to this point, the teacher may decide to release question 3 to students or continue modeling to ensure better understanding of the new content before releasing in the next component.

3.

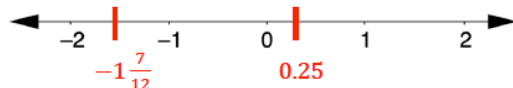
- a. On the number line, make a tick mark that is halfway between each value on the number line. Label each tick mark with the value of its location.



- b. Make another tick mark halfway between each value and the tick marks that were previously made. Label each with the value of its location.



- c. Plot the values $-1\frac{7}{12}$ and 0.25 on the number line.



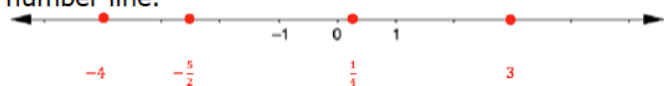
3.1.6 Try It: Plotting Rational Numbers on a Number Line

Component Overview: Students, independently or with a partner, practice plotting integers, fractions, and decimals on horizontal and vertical number lines. Circulate while students work to support struggling students, monitor progress, and select which students you wish to have share during the debrief.

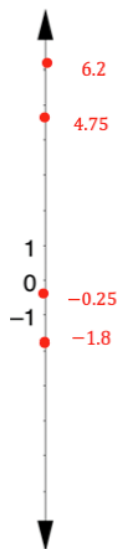


3.1.6 Try it: Plotting Rational Numbers on a Number Line

- Plot the numbers $-\frac{5}{2}$, -4 , $\frac{1}{4}$, and 3 on the horizontal number line.



- Plot the rational numbers -0.25 , -1.8 , 6.2 , and 4.75 on the vertical number line.



What type of numbers are below/to the left of zero? What type of numbers are above/to the right of zero?

3.1.7 Your Turn!

Component Overview: Students independently practice plotting integers, fractions, and decimals on horizontal and vertical number lines. Students are provided with a visual entry point through the use of number lines. If time is an issue, then consider assigning 3.1.7 Your Turn! for homework.



3.1.7 Your Turn!

- Negative numbers are ☐ greater ☒ less than zero. On a horizontal number line, negative numbers are to the ☐ right ☒ left of zero. On a vertical number line, negative numbers are ☐ above ☒ below zero.

- Plot the integers 2, -3, 0, -5 and 5 on the horizontal number line.



- Plot the rational numbers $-1\frac{1}{5}$, $-\frac{4}{5}$, 2, and $\frac{6}{5}$ on the vertical number line.



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

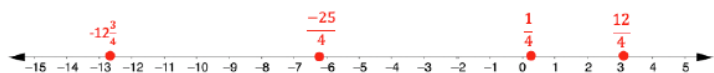
3.1.8 Wrap-Up: Ticket Out the Door

Component Overview: In this formative assessment, students demonstrate their understanding of plotting rational numbers on a number line. Use information from 3.1.8 Wrap-Up to decide which students need remediation before the next lesson.

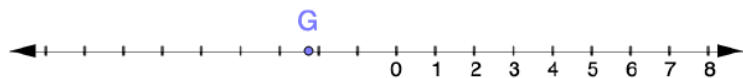


3.1.8 Wrap-Up: Ticket Out the Door

- Plot the rational numbers $-\frac{25}{4}$, $-12\frac{3}{4}$, $\frac{12}{4}$, and $\frac{1}{4}$ on the number line.



- Explain how to estimate the value of G .



Answers will vary. G is to the left of zero, so it will be a negative value. It's between -2 and -3 , but closer to -2 , so I would estimate its value is about -2.25 .

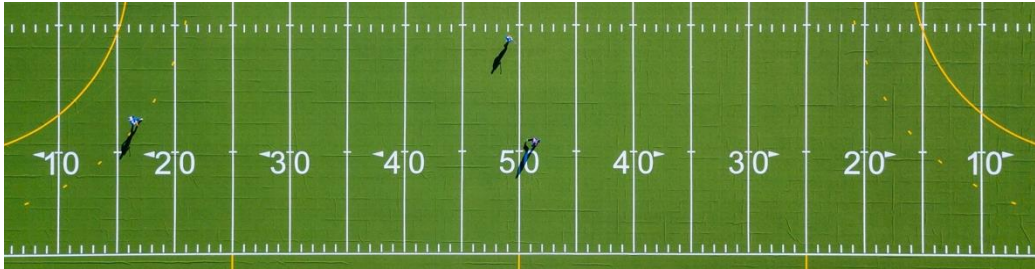


How can converting fractions to decimals or vice versa help you determine where to place rational numbers on a number line?

3.2 Rational Numbers and the Number Line – Part 2

Lesson Introduction

 Benchmark	MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.		 Learning Targets	- I can determine how far a number and its opposite are from zero. - I can represent a number and its opposite on a number line. - I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within context.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- What is the meaning of a number’s opposite? - How can you use tick marks between each value to determine the locations of rational numbers on a number line? - How is zero used to determine where to plot a number and its opposite on a number line?			
 Lesson Narrative	In this lesson, students will work on developing their understanding of a number and its opposite. During the lesson, students will use a number line to represent a number and its opposite, showing that they are the same distance from zero on a number line. Students also use the number line to compare numbers and their opposites.			
 Component Summary	3.2.1 Warm-Up (~5 min.)	Notice and wonder about words that are opposites (<i>SE p. 82</i>)	3.2.4 Try It (5 min.)	Plot opposites on horizontal and vertical number lines (<i>SE p. 84</i>)
	3.2.2 Activity (10 min.)	Explore the relationship between opposites on number lines (<i>SE p. 83</i>)	3.2.5 Guided Instruction (5 min.)	Compare numbers and their opposites using greater and less than symbols (<i>SE p. 85</i>)
	3.2.3 Collaboration (10 min.)	Plot numbers and their opposites on horizontal number lines (<i>SE p. 83</i>)	3.2.6 Your Turn! (5-10 min.)	Plot values on a number line and compare their values using greater and less than symbols (<i>SE p. 86</i>)
	3.2.7 Wrap-Up (~5 min.)	Describe the relationship between opposites and zero on a number line (<i>SE p. 86</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational numbers - Opposites		- Paper to create foldable - Number line from Lesson 1	
			Additional Preparation	
			(Optional) For 3.2.2 Activity, prepare labeled number lines to save time in the lesson.	

 Teacher Tips	Common Misconceptions - Students may struggle with plotting negative fractions and decimals when using irregular intervals, particularly in 3.2.4 Try It. - Students may get rational numbers and integers confused in 3.2.3 Collaboration. - Students may think inequality symbols can only be read in one direction. For example, they may not understand $a < b$ can be interpreted as “a is less than b,” and also as “b is greater than a.”	Things to Consider - Use the number line created to show that when folded at zero, opposite numbers will overlap, showing that they are the same distance from zero. - Use different colors to help students find the distance from zero. - In 3.2.2 Activity, describe what a vertical crease is. - Consider using graph or lined paper to create the foldable number lines to ensure equally spaced tick marks.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 3.2.1 Warm-Up, students notice and wonder about familiar words that are opposites. This activity develops students’ conceptual understanding of opposite numbers, as they are able to connect the new concept to vocabulary terms they may already be familiar with.	MA.K12.MTR.4.1: Discussion and Reflection In 3.2.3 Collaboration, students analyze and critique the thinking of others. They articulate their thinking as they discuss their analysis of the samples of student work.
 Differentiate Instruction	3.2.2 Activity provides kinesthetic and visual entry points as students continue developing their understanding of negative values. Get creative and use real-world examples like a football field or the human wingspan to engage students.	
		
Lesson Notes:		

3.2.1 Warm-Up: Word Pairs

Component Overview: Students develop a conceptual understanding of opposites by connecting the concept of opposite to real-world vocabulary they may be familiar with. Students are asked what they notice and wonder about the pairs of words.



3.2.1 Warm-Up: Word Pairs

Up	Down
Left	Right
Clockwise	Counterclockwise
Increase	Decrease
Ebb	Flow
West	East
North	South
Gain	Lose



This activity can be facilitated as a Notice and Wonder routine. Students do not need to identify the pairs of words as opposites, but if they do, use it as a bridge to the math application of opposites.

1. What do you notice about the pairs of words above?

Answers will vary. They are opposites. They involve direction.

2. How do you think this may be used in the study of mathematics?

Answers will vary. Opposites could be represented with positive and negative values.

3.2.2 Activity: Folded Number Line

Component Overview: Students explore opposites on a number line. If using a sheet of paper as a foldable, ensure students crease the paper exactly on the zero of their number lines. After folding, have students discuss the relationship between numbers and their opposites relative to zero on the number line.



3.2.2 Activity: Folded Number Line

On a separate sheet of paper, create a number line with integers from -10 to 10 to answer the following questions. Fold the number line so that the vertical crease goes through 0 and the two sides of the number line match up perfectly.

- When the number line is folded, which number is the same distance away from 0 as the number 4 ?
 -4
- When the number line is folded, which number is the same distance away from 0 as the number -7 ?
 7
- When the number line is folded, what do you notice about the numbers that are the same distance away from 0 ?
Answers will vary. They are the same number, but one is positive and one is negative.
- Find another pair of numbers that are the same distance away from 0 .
Answers will vary. 1 and -1 . 9 and -9 .
- Two numbers that are the same distance from zero on the number line are called opposites.



If you have markers that bleed, consider having students draw and label only half of the number line or only the positive numbers. Then fold in half. If the marker bleeds well, the students will unfold to reveal a mirror image of the number line. Have students then correctly label those values as negative, identify opposites, and describe their relationship in terms of distance from zero.



What do you notice about the location of opposites on a number line? Is this always true? Why or why not?

3.2.3 Collaboration: Opposites

Component Overview: Students choose their own two values, one positive and one negative, for W and F . Students will therefore have a variety of answers. This activity is intentionally scaffolded, with each question acting as a natural question stem and next step for understanding opposites. Students should complete the entire activity with their partner before debriefing any answers.



3.2.3 Collaboration: Opposites

With your partner, choose values for W and F . The value for W can be any rational number between -8 and 0 that is not an integer. The value for F can be any rational number between 0 and 8 that is not an integer.

Answers will vary.

The value for W is -5.2 . The value for F is 3.5 .

- Plot and label both numbers on the number line.

Answers will vary based on choice of W and F .



- Plot the opposite of W . Label the point X .

Answers will vary based on choice of W . Answer shown on number line above.

- What is the value of point X ?

Answers will vary based on choice of W . 5.2

- How far is point X from 0 ?

Answers will vary based on choice of W . 5.2 units

- Plot the opposite of F . Label the point G .

Answers will vary based on choice of F . Answer shown on number line above.

- What is the value of point G ?

Answers will vary based on choice of F . -3.5

- How far is point G from 0 ?

Answers will vary based on choice of F . 3.5 units

- Noah said, "The expression $-a$ can be read 'negative a ' or 'the opposite of a '. So, if a is a rational number, $-a$ will always be a negative number."

Use points F , G , W , and X to explain to Noah that the opposite of a value could be positive.

Answers will vary. The opposite of a positive number will always be negative. The opposite of a negative number will always be positive. The expression " $-G$ " can be read "negative G " or "the opposite of -3.5 " which is positive 3.5.



This activity is intentionally scaffolded to guide students' thought processes for discovering a number and its opposite's relationship, relative to their distance from zero on a number line. Review the 3.2.7 Wrap-Up question and determine framing questions to use here to best prepare students for the Wrap-Up.



Is Noah correct? Why or why not? Can you have an opposite of an opposite? What would that look like represented mathematically? Does "opposite" always mean "negative"?

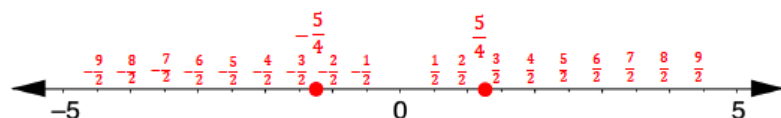
3.2.4 Try It: Opposites

Component Overview: Students apply skills of plotting opposites on both horizontal and vertical number lines. Students compare values, identifying which is larger. Students should recognize that positive values are always greater in value than negative values.



3.2.4 Try It: Opposites

- Use the number line below to answer the following problems.



- Label the tick marks on the number line. Then plot the value $\frac{5}{4}$ and its opposite on the number line.
- Which value is larger, $\frac{5}{4}$ or its opposite?

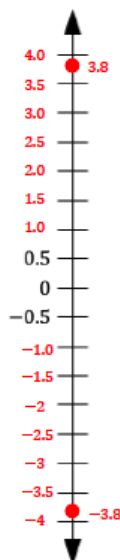
$\frac{5}{4}$

- Use the vertical number line to answer the following problems.

- Label the tick marks on the number line. Plot and label the value -3.8 and its opposite on the number line.

- Which value is larger, -3.8 or its opposite?

3.8 is larger.



Consider using a Think Aloud for a small group of students that need additional support.

- How do you determine how to label a number line? What scale do you use?
- What is the range of the part you are labeling ($0 - 5$)?
- How many tick marks are there (10)?
- So how could you divide five into 10 equal parts?
- What would the value of each part be if 10 equal parts equaled five? ($\frac{1}{2}$ or 0.5)



Students do not have to label every tick mark if they do not need to do so to determine the location of each value on the number lines. For example, they could label only the integer values, if they wish, since this would not affect the placement of the points on the number lines.

3.2.5 Guided Instruction: Comparing a Quantity and Its Opposite

Component Overview: Students review how to use less than and greater than symbols to compare two values. Students finish the activity by reasoning about the values of numbers and their opposites, relative to their position on a number line and whether they are positive or negative.

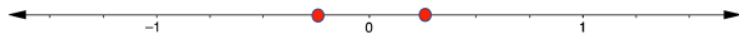


3.2.5 Guided Instruction: Comparing a Quantity and its Opposite

Plot each pair of values on the number line. Choose an answer to correctly fill in the blank.

1.

- a. Plot -0.25 and 0.25 .

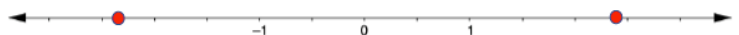


- b. The value -0.25 is ☐ greater ☒ less than 0.25 .

- c. -0.25 ☐ $>$ ☒ $<$ 0.25

2.

- a. Plot $2\frac{2}{5}$ and $-2\frac{2}{5}$.



- b. The value $2\frac{2}{5}$ is ☐ greater ☒ less than $-2\frac{2}{5}$.

- c. $2\frac{2}{5}$ ☐ $>$ ☒ $<$ $-2\frac{2}{5}$

3. In each comparison, the number on a horizontal number line that is farther to the ☐ right ☒ left is the greater value.

4. When comparing opposites, the ☐ positive ☒ negative value will always be the greater value.



Have students practice interpreting these mathematical statements, particularly reading the same statement in multiple ways. For example, “ -0.25 is less than 0.25 , and 0.25 is greater than -0.25 .” This will support students’ conceptual understanding of inequality statements.

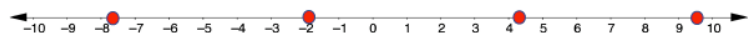
3.2.6 Your Turn!

Component Overview: Students continue practicing plotting, ordering, and comparing values of rational numbers. Circulate while students work to support struggling students, monitor student progress, and select students that you want to have share during the debrief.



3.2.6 Your Turn!

- Plot and label each value and its opposite on the number line: 9.5, -2, 4.2, $-7\frac{3}{4}$.



- Pairs of opposite numbers are ☐ the same ☐ different distance from zero on a number line.
- Complete each comparison statement for the pairs of numbers you plotted in problem #1. For each statement, use < or > in the first blank and the opposite of the given number in the second blank.

a. 9.5 > -9.5

b. -2 < 2

c. 4.2 > -4.2

d. $-7\frac{3}{4}$ < $7\frac{3}{4}$



For students who might finish early, consider challenging them by beginning to compare values other than opposites. In the following lessons, students begin comparing the values of rational numbers.



What is your strategy for plotting fractions and decimals? How do you determine which integers to plot it between and how close or far to plot from an integer?



Consider providing guiding questions for students who struggle with plotting decimals and fractions. For example, if plotting 4.2, students can ask themselves questions like:

- What is the closest whole number?
- Is it greater or less than that number?
- Does that mean it will be to the right or left of it on the number line? How far away?

3.2.7 Wrap-Up: Cool Down

Component Overview: This Wrap-Up evaluates students' understanding of opposites and their locations on the number line.



3.2.7 Wrap-Up: Cool Down

1. What is the relationship between a number and its opposite? Use the words **number line** and **zero** in your explanation.

Answers will vary. On a number line, a value and its opposite are the same distance from zero, but on opposite sides of zero. For example, the opposite of 4 is -4 . Both 4 and its opposite, -4 , are 4 units away from zero on a number line.

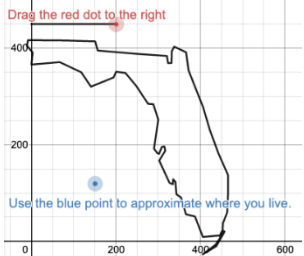


For students who might struggle with communicating their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.

3.3 Plotting, Ordering, and Comparing Rational Numbers – Part 1

Lesson Introduction

 Benchmark	MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.		 Learning Target	- I can plot and compare rational numbers represented as a decimal or percent. - I can plot, compare, and order fractional representations of rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.		MA.8.NSO.1.2 Plot, order and compare rational and irrational numbers, represented in various forms.	
 Essential Questions	- How can number lines be used to determine which value(s) are greater or less than others? - How can statements of inequality comparing rational numbers be interpreted in context? - How do you plot rational numbers on number lines? - How do you compare rational numbers?			
 Lesson Narrative	Students are introduced to plotting and comparing (using $<$, $>$, or $=$) rational numbers represented as decimals and percentages. The lesson begins incorporating some real-world contexts but focuses more on the mathematical context. Students will use number lines, extending their understanding from Lessons 1-2, to compare values using comparison symbols. Connections will also be made to students' prior understanding from Unit 2, where they learned to rewrite positive rational numbers in different forms.			
 Component Summary	3.3.1 Warm-Up (~5 min.)	Notice and Wonder with temperatures in two cities (<i>SE p. 87</i>)	3.3.3 Exploration (10 min.)	Explore how to plot and compare fractions (<i>SE p. 90</i>)
	3.3.2 Guided Instruction (10-15 min.)	Guided learning of comparing and ordering decimals and percentages in context (<i>SE p. 88</i>)	3.3.4 Your Turn! (10-15 min.)	Practice plotting, ordering, and comparing rational numbers (<i>SE p. 91</i>)
	3.3.5 Wrap-Up (~5 min.)	Self-assessment of understanding of lesson's learning targets (<i>SE p. 92</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational number - Decimal - Percent - Inequality symbol		N/A	(Optional) Find maps and pictures to help students understand the temperatures in different locations in 3.3.1 Warm-Up and 3.3.2 Guided Instruction. - Consider providing prepared number lines with $\frac{1}{2}$ and/or $\frac{1}{4}$ tick marks.

	Common Misconceptions	Things to Consider
 Teacher Tips	- Students may not initially recognize that numbers farther to the left on the number line are smaller values (and vice versa). - Students may try to use shortcuts (i.e., moving the decimal a specified number of places) when converting values and making comparisons. - Students may struggle interpreting statements of inequality in context, particularly as they relate to temperatures and race times. - Students may struggle when comparing and ordering fractions if they do not recall how to rewrite fractions with common denominators. - Students may struggle when comparing and ordering decimals if they do not recall how to line up the place values.	- There is a number line within 3.3.2 Guided Instruction to provide an opportunity to make the connection that numbers farther to the left are smaller values. - Use the language “is less than” or “is greater than” rather than “less than” or “greater than” when reading inequality statements. Using the latter can create misconceptions later when students interpret verbal statements and relate them to algebraic statements. For example, “3 less than x” would be expressed as “$x - 3$” not “$3 < x$.” If this is not modeled correctly when using inequalities, it will be hard to undo later.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Compare and Connect 3.3.3 Exploration provides opportunity for students to discuss different strategies for comparing, ordering, and plotting rational numbers. Facilitate discussions to help students connect new strategies to prior knowledge.	MA.K12.MTR.1.1: Participate in Effortful Learning Students have opportunities to actively participate in effortful learning individually in 3.3.4 Your Turn! Students must use higher-order thinking skills as they plot, order, and compare decimals. Students can work independently and then with a partner or group to encourage and support active problem-solving. 
Lesson Notes:		

3.3.1 Warm-Up: Notice and Wonder

Component Overview: Students share what they notice and wonder about the temperatures in two different cities to apply their understanding of positive and negative numbers from previous lessons.



3.3.1 Warm-Up: Notice and Wonder

- The diagram gives the temperature, in Fahrenheit ($^{\circ}\text{F}$) and in Celsius ($^{\circ}\text{C}$), for Memphis, Tennessee, and Bangor, Maine.



What do you notice about the temperatures?

*Answers will vary. Memphis has positive values only.
Bangor has both positive and negative values.*



*Can you create a number line to compare temperatures?
Students can create a vertical number line to display the
Memphis and Bangor temperatures.*

Consider this extension for homework.

3.3.2 Guided Instruction: Comparing and Ordering Decimals

Component Overview: Students compare and order decimals in the context of temperature before extending this to compare decimals and percentages in the context of population.

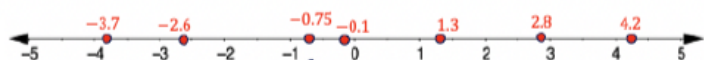


3.3.2 Guided Instruction: Comparing and Ordering Decimals

1. The table shows the lowest temperature, in degrees Celsius ($^{\circ}\text{C}$), for each day during the week of October 14, 2019 in Anchorage, Alaska.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temperature ($^{\circ}\text{C}$)	4.2	-3.7	-0.75	1.3	2.8	-0.1	-2.6

- a. Plot the temperatures on the number line.



- b. Which temperature did you plot farthest to the left?
-3.7
- c. What day of the week is represented by the value that was plotted farthest to the left?
Tuesday
- d. Why does the location of the temperature that is plotted farthest to the left help determine which day had the lowermost temperature?
Answers will vary. Values decrease as you move to the left on the number line.
- e. Interpret the temperature in this context.
*During the week of October 14, 2019, the lowermost temperature was **-3.7** $^{\circ}\text{C}$. This was the*

☐ warmest
☒ coldest

temperature reached that week.
- f. The inequality symbols, $<$ or $>$, can be used to compare two of the daily low temperatures. Write an inequality that shows the relationship between two of the low temperatures for the week of October 14, 2019.

Answers will vary. Example: $-3.7 < -2.6$



Consider showing students the geographical location of states referenced in this component (Alaska, Florida, etc.) on a map and discuss potential differences in the climate.



How can we use a number line to help us determine which value(s) are greater than others? Does this value lie to the left or right of zero on a horizontal number line? Would a vertical number line make sense here? Why or why not?

3.3.2 Guided Instruction: Comparing and Ordering Decimals (continued)

2. Jayden, a data analyst, is comparing the populations of four different states to the population of the United States. His data is shown in the table.

State	Population as a Portion of the U.S. Population (decimal form)	Population as a Percentage of the U.S. Population (percent form)
California	0.119	
Florida		6.4%
New York		5.9%
Texas	0.087	

- a. Choose two states to compare. Write or describe your comparison.

Answers will vary. Example: I can compare populations of California and Texas using decimal numbers. Since $0.119 > 0.087$, California has a larger population than Texas.

Example: Florida has a larger population than New York because 6.4% of the US population is more than 5.9%.

- b. Olivia and Anthony's strategies for comparing California and New York are explained below.

Olivia explained, "I know that percent represents the part per 100, so the population of New York is approximately $\frac{6}{100}$ of the population of the U.S. The portion of the U.S. population living in California is approximately 12 hundredths, or $\frac{12}{100}$. Therefore, more people live in California than in New York."

Anthony explained, "I can convert 5.9% to a decimal by dividing 5.9 by 100. So 5.9% can also be represented by 0.059. I know 0.119 is greater than 0.059. Therefore, more people live in California than in New York."

3.3.2 Guided Instruction: Comparing and Ordering Decimals (continued)

Olivia and Anthony both used different methods to come to the same conclusion. Discuss their methods with a partner.

Olivia's Method	Anthony's Method
<i>Answers will vary.</i> <i>Olivia converted the decimal to a percentage by rewriting it as a fraction out of 100 first.</i>	<i>Answers will vary.</i> <i>Anthony converted the percentage to a decimal by dividing the percent by 100 so that the values could be compared.</i>

- c. Complete each inequality statement below to compare the given values.

$$0.119 > 6.4\% \quad 6.4\% < 0.087 \quad 0.087 > 5.9\%$$

- d. In which of the four states does the highest portion of the U.S. population live?
California
- e. In which of the four states does the smallest portion of the U.S. population live?
New York



Students can role play each scenario, reading each student's response as their own. This strategy provides auditory entry points for students who prefer learning with verbal processing.



Consider prompting students to rewrite one of the values in each inequality so that both values are in the same form as an additional scaffold. Model how to do this for the first inequality before releasing the others to students.

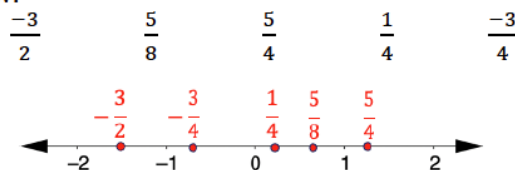
3.3.3 Exploration: Plotting and Comparing Fractions

Component Overview: Students explore how to plot and compare positive and negative fractions using a number line. Circulate while students work to monitor progress, assist struggling students, and select students to share their strategies during the whole-class debrief.



3.3.3 Exploration: Plotting and Comparing Fractions

- Plot and label the rational numbers on the number line below.



Using the rational numbers plotted above, answer the following questions.

- Andre says that $\frac{1}{4}$ is less than $-\frac{3}{4}$ because out of the two values, $\frac{1}{4}$ is closer to 0. Explain why you agree or disagree with Andre.

Answers will vary. I disagree because $-\frac{3}{4}$ is negative, so it is less than $\frac{1}{4}$ because all negative numbers are smaller than positive numbers.

- Complete the table for each pair of numbers.

Pair	Farthest Distance From 0	Greater Value
$\frac{1}{4}$ or $\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$
$-\frac{3}{4}$ or $\frac{5}{8}$	$-\frac{3}{4}$	$\frac{5}{8}$

- Is the value that is farther from 0 always the greater value? Explain your reasoning.

Answers will vary. No, the value that is farther from 0 is not always the greater value because it depends upon the sign of that value.



Compare and Connect

Prompt students to compare and connect their prior knowledge of plotting and comparing integers and positive fractions to plotting and comparing positive and negative fractions.



Consider providing digital number lines for students to verify their answers as they explore.

3.3.4 Your Turn!

Component Overview: Students practice plotting and comparing rational numbers.



3.3.4 Your Turn!

1. Explain why each inequality statement is true or false.
a. $-5.2 < -7.1$

Answers will vary. This is false because both values are negative, but -7.1 is farther to the left on the number line than -5.2 , so it is smaller.

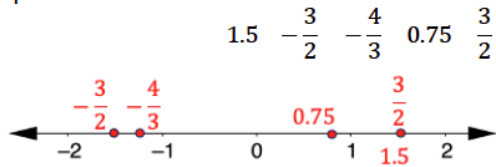
- b. $3.1 > 38\%$

Answers will vary. This is true because $3.1 = 310\%$ or $38\% = 0.38$. $3.1 > 0.38$ and $310\% > 38\%$.

2. Use the symbols $<$, $>$ or $=$ to compare each pair of values.

$$-\frac{4}{3} > -\frac{3}{2} \quad 0.75 < \frac{3}{2} \quad \frac{3}{2} = 1.5$$

3. Plot the following values on the number line. Label each point with its numeric value.



- a. Complete the inequality statement to show the relationship between the values.

$$-\frac{3}{2} > -\frac{4}{3} > 0.75 > 1.5$$

4. Select all the values that are greater than -5 .

- ☒ 1.324
☐ -6.2
☐ -12.67
☒ 0.231
☐ -1.4
☒ -4.789

5. Order these values from least to greatest.

0.43, -4.2 , 0, 50%, -1.5 , -6.712 , 19%
 -6.712 , -4.2 , -1.5 , 0, 19%, 0.43, 50%



Between which two integers does the value lie? To which integer is it closest? How will converting rational numbers in different forms to the same form help me to compare and order two or more numbers accurately?



Prompt students to create their own real-world problems that require the ordering and/or comparing of rational numbers. Have students trade and solve each other's problems.

3.3.5 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment of their understanding of the lesson's learning targets using an emoji scale. Then students provide a written explanation to support their choice of emoji.



3.3.5 Wrap-Up: Reflection

1. Circle the emoji that best represents your understanding of the lesson today.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10



Consider grouping students for collaboration in Lesson 4 based on their response. Alternatively, consider placing a number line on chart paper for students to mark as they walk out of the door for a visual spread.

2. Write an explanation of why you chose that emoji.

Answers will vary.

3.4 Plotting, Ordering, and Comparing Rational Numbers – Part 2

Lesson Introduction

 Benchmark	MA.6.NSO.1.1 Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers.		 Learning Target	I can plot, compare, and order rational numbers.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.NSO.1.4 Plot, order, and compare multidigit numbers with decimals up to the thousandths.		MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real-number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.	
 Essential Questions	- How do you convert a decimal to a fraction and a fraction to a decimal? - How can you compare rational numbers using a number line?			
 Lesson Narrative	Students extend their learning to plot, compare, and order rational numbers. Students plot rational numbers on a number line. They use the numbers' placement on the number line to order and compare the numbers. Students convert fractions to decimals, and vice versa, to aid in comparing and ordering rational numbers. Students only compare negative values expressed in the same form for specified intervals along the number line. Positive values are compared in any form.			
 Component Summary	3.4.1 Warm-Up (~5 min.)	Error analysis for plotting negative values on a number line (<i>SE p. 93</i>)	3.4.4 Guided Instruction (5-10 min.)	Compare rational numbers and analyze different methods of comparing rational numbers (<i>SE p. 95</i>)
	3.4.2 Activity (10 min.)	Place numbers on a life-sized number line (<i>SE p. 94</i>)	3.4.5 Try It (5 min.)	Practice plotting, ordering, and comparing rational numbers (<i>SE p. 96</i>)
	3.4.3 Exploration (10 min.)	Compare unknown numbers based on their placement on a number line (<i>SE p. 94</i>)	3.4.6 Your Turn! (5 min.)	Additional practice plotting, ordering, and comparing rational numbers (<i>SE p. 97</i>)
	3.4.7 Wrap-Up (~5 min.)	Demonstrate understanding of plotting, ordering, and comparing rational numbers (<i>SE p. 97</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Rational number - Less/Greater than - Equivalent - Opposite		- Class number line - Sticky notes	
			Additional Preparation	
			(Optional) Print and cut out the set of rational number cards if you do not plan to use sticky notes for 3.4.2 Activity.	

 Teacher Tips	Common Misconceptions - Students may interpret larger denominators as larger in value. - Students may interpret decreasing negative values as getting larger in values as the number gets “bigger.” - Students may believe inequality symbols can only be read left to right.	Things to Consider - In 3.4.4 Guided Instruction, prompt students to reflect on the different strategies they learn for comparing and ordering rational numbers written in different forms. - Consider using the Think Aloud strategy when demonstrating how to determine the location of a rational number on the number line.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display In 3.4.3 Exploration, circulate while students are working with their partners. Collect different answers, strategies, and justifications. Debrief the activity by displaying different students’ work. Discuss the students’ responses.	MA.K12.MTR.2.1: Multiple Representations Students represent rational numbers as decimals, percentages, and fractions, (i.e., 0.4, 40%, and $\frac{2}{5}$). Students use equivalent forms to compare and order the rational numbers in 3.4.6 Your Turn!
 Differentiate Instruction	Consider utilizing On-Ramp to 6th Grade as an additional tool for scaffolding content related to conversions between fractions and decimals and percentages. - Number Lines: Mixed Numbers - Number Lines: Fractions - Representing Fractions on a Number Line - Compare and Order Decimals with Models - Compare and Order Decimals with $>$, $=$, or $<$ - Comparing Fractions Using Symbols $>$, $=$, or $<$ - Comparing Fractions with Like Numerators and Denominators - Compare and Order Whole Numbers with $>$, $=$, or $<$	
Lesson Notes:		

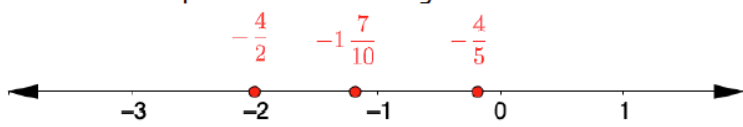
3.4.1 Warm-Up: Find the Error

Component Overview: Students analyze a sample of student work. They determine which values are plotted correctly and which are not. Focus on asking students to justify their answers to check for understanding.



3.4.1 Warm-Up: Find the Error

1. A student plotted the following numbers.



2. Which values are plotted correctly?

$$-\frac{4}{2}$$

3. Which values are NOT plotted correctly?

$$-1\frac{7}{10} \quad -\frac{4}{5}$$



- What feedback would you give to a student who made the errors you noticed?
- How can we use integers to estimate the location of fractions on a number line?
- How do you know $-1\frac{7}{10}$ should be closer to -2 than to -1 ? Could we say the same thing for $-\frac{4}{5}$? Why or why not?

3.4.2 Activity: Class Number Line

Component Overview: Construct a class number line on a wall in your classroom, the floor, or even a virtual line projected onto the board using a platform like Desmos. You can either write the numbers on the notes ahead of time or simply provide a list of numbers that students can then select and write on their sticky notes.



3.4.2 Activity: Class Number Line

Your teacher will give you a sticky note with a rational number written on the front.

Write your name on the back of the sticky note and place your number in the correct location on the class number line.

Class Number Line Values (ordered from least to greatest)

$-4\frac{3}{4}$	$-4\frac{7}{10}$	$-4\frac{1}{2}$	$-4\frac{1}{5}$	$-4\frac{1}{10}$
-3.9	-3.75	-3.7	-3.25	-3.1
$-2\frac{97}{100}$	$-2\frac{1}{5}$	$-\frac{4}{2}$	$-1\frac{7}{10}$	$-\frac{4}{5}$
$\frac{1}{4}$	36%	0.6	78%	$\frac{91}{100}$
$1\frac{1}{5}$	125%	$\frac{3}{2}$	1.9	2.01
$2\frac{1}{10}$	$2\frac{3}{5}$	2.75	3.25	$3\frac{3}{10}$



The values can either be written on sticky notes or cards could be used with tape.



Consider your students when assigning the values or provide them with an opportunity to select their own value.

Consider also chunking this activity so that groups of 5 are formed and have them decide how to order within their group first, before placing their values on the class number line.



Probe students to share their reasoning.

- How did you determine where to place your number?
- What integer is your number closest to?
- How do you determine where to plot decimals and fractions?
- Did the placement another student made for his/her value help you determine where to place yours? Explain.

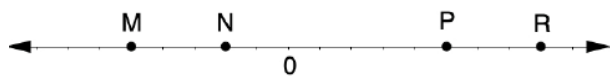
3.4.3 Exploration: Comparing Points on a Number Line

Component Overview: Students compare unknown numbers based on their placement on a number line. Circulate while students are working with their partners. Collect different answers, strategies, and justifications. Debrief the activity by displaying different students' work. Discuss the students' responses.



3.4.3 Exploration: Comparing Points on a Number Line

A number line is shown with points M, N, P , and R . Work with your partner to answer the questions about the number line.



- Use each of the given phrases to describe or compare the values of points M, N, P, R . Each statement will use one or two of the points.

Phrase	Description or Comparison of M, N, P or R
less than	<i>Answers will vary. M is less than N.</i>
opposite of	<i>Answers will vary. M is the opposite of P.</i>
greater than	<i>Answers will vary. R is greater than P.</i>
negative number	<i>Answers will vary. M is a negative number.</i>

- If P is $2\frac{1}{2}$, what are the values of M, N , and R ?
 M is $-2\frac{1}{2}$, N is -1 , R is 4
- If N is -0.4 , what are the values of M, P , and R ?
 M is -1 , P is 1 , R is 1.6
- If R is 200% , what are the values of M, N , and P ?
 M is -125% , N is -50% , P is 125%
- If M is $-1\frac{2}{3}$, what are the values of N, P , and R ?
 N is $-\frac{2}{3}$, P is $1\frac{2}{3}$, R is $2\frac{2}{3}$



Think-Pair-Share

Depending on the needs of your students, consider giving students the opportunity to engage independently first and refine their responses with a partner before sharing.

3.4.4 Guided Instruction: Comparing Rational Numbers in Different Forms

Component Overview: Students compare fractions, decimals, and percentages using inequality symbols. Students analyze different methods of comparing rational numbers, including using a number line, finding a common denominator, and rewriting in an equivalent form.



3.4.4 Guided Instruction: Comparing Rational Numbers in Different Forms

- Use the symbols $>$, $<$ or $=$ to compare each pair of values.

$$9.02 > 9.2\% \quad -\frac{3}{5} > -\frac{13}{20} \quad 6.050 = 6.05$$

$$0.4 > \frac{9}{40} \quad 2\frac{3}{4} > 2.624 \quad -4\frac{1}{3} < -\frac{10}{3}$$

- Emma and Noah shared their strategy for comparing values written as fractions. Write your strategy in the box and note any **differences** from Emma's or Noah's strategy.

Emma's Strategy	Noah's Strategy
When Emma compares a negative fraction to a negative fraction, she starts by thinking about which whole numbers each fraction lies between and which each is closest to.	When Noah compares a negative fraction to a negative fraction, he uses common denominators.
<i>Answers will vary. I converted fractions into decimals to compare the values. My strategy is different from Emma's because she estimated, and my answer is different from Noah's because he did not use decimals.</i>	

- Angel and Sophia share their strategies for comparing a value written as a decimal to a value written as a fraction. Write your strategy in the box and note any **similarities** from Angel's or Sophia's strategy.

Angel's strategy	Sophia's strategy
When Angel compares a positive decimal to a positive fraction, he rewrites the fraction as an equivalent decimal.	When Sophia compares a positive decimal to a positive fraction, she plots them on a number line.
<i>Answers will vary. I converted fractions into decimals to compare the values, so my strategy was the same as Angel's.</i>	



There is more practice in the next activity, so spend more time on reasoning and verbalizing strategies than finding answers (process over product).



Consider having all students read all four strategies at once and set up a Four Corners named for each student. Students will read all four strategies, then walk to a corner based on the approach or strategy they align with most. At each corner, students explain why they chose that corner.

3.4.5 Try It: Plotting, Ordering, and Comparing Rational Numbers in Different Forms

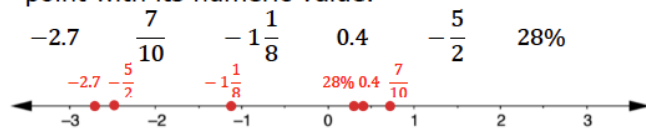
Component Overview: Students plot, order, and compare rational numbers in a variety of forms. The questions include fractions, decimals, and percentages. Encourage students to choose one or more strategies based on the previous component.



3.4.5 Try It: Plotting, Ordering, and Comparing Rational numbers in Different Forms

1.

- a. Plot each value on the number line. Label each point with its numeric value.



- b. What strategies did you use that were helpful when plotting the rational numbers written in different forms on the number line?

Answers will vary. I converted all the numbers to decimals to help me plot them on the number line.

- c. Order the values from greatest to least.

$\frac{7}{10}$ 0.4 28% $-1\frac{1}{8}$ $-\frac{5}{2}$ -2.7

- d. Complete the inequality statement to show the relationship between the values plotted on the number line.

$$-2.7 < -\frac{5}{2} < -1\frac{1}{8} < 28\% < 0.4 < \frac{7}{10}$$



How does the order of values from greatest to least relate to the location on a number line? Is this always true? Why or why not?

3.4.6 Your Turn!

Component Overview: Students continue practicing plotting, ordering, and comparing rational numbers.

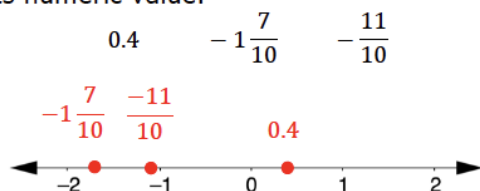


3.4.6 Your Turn!

1. Order the values from least to greatest.

$\frac{1}{2}$ 0 1 $-1\frac{1}{2}$ $-\frac{1}{2}$ -1
 $-1\frac{1}{2}$ -1 $-\frac{1}{2}$ 0 $\frac{1}{2}$ 1

2. Plot the values on the number line. Label each point with its numeric value.



3. Order the values from greatest to least.

-2 0 $-\frac{1}{6}$ $1\frac{3}{8}$ $-3\frac{1}{3}$ 1.4 $\frac{5}{4}$ $-\frac{13}{10}$ $\frac{21}{5}$
 $\frac{21}{5}$ 1.4 $1\frac{3}{8}$ $\frac{5}{4}$ 0 $-\frac{1}{6}$ $-\frac{13}{10}$ -2 $-3\frac{1}{3}$

4. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

$8.7 > 8.689$ $-\frac{5}{3} < -\frac{7}{10}$ $7.2 < 7\frac{1}{3}$
 $0.38 > 5\%$ $-4\frac{5}{8} < -4.5$ $-3\frac{2}{5} = -3.4$



Students might still struggle interpreting negative values or fractions in order to compare. Consider what question stems you can provide for students to use when they get stuck.

- What whole number is this value closest to?
- Is it greater than or less than that whole number?
- How do you know? Give an example.
- Should that value be plotted on the left or the right of that whole number?
- Is that value greater or less than...?

3.4.7 Wrap-Up: Ticket Out the Door

Component Overview: In this formative assessment, students demonstrate their ability to plot, order, and compare rational numbers. Use the data gathered to make instructional decisions regarding remediation prior to the next lesson.



3.4.7 Wrap-Up: Ticket Out the Door

1. Use the values 3.6 , $-4\frac{4}{5}$, 0.7 , $-1\frac{1}{5}$, $\frac{15}{4}$, 89% , and $-\frac{7}{8}$. Plot the values on the number line. Label each point with its numeric value.



By plotting values in question, students must use skills of comparing and ordering they would need in question 2. Question 3 will be important in gathering data for students' understanding of inequality symbols.

2. List the values 3.6 , $-4\frac{4}{5}$, 0.7 , $-1\frac{1}{5}$, $\frac{15}{4}$, 89% , and $-\frac{7}{8}$ from least to greatest.

$-4\frac{4}{5}$ $-1\frac{1}{5}$ $-\frac{7}{8}$ 0.7 89% 3.6 $\frac{15}{4}$

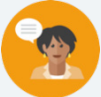
3. Use the symbols $<$, $>$, or $=$ to compare each pair of values.

$-12.742 > -12.9$ $1\frac{2}{9} < 150\%$ $3\frac{7}{12} < 3.7$

3.5 Rational Numbers in Real-World Context

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite directions using rational numbers. Compare them on a number line and explain the meaning of zero within its context.		 Learning Targets	- I can determine positive and negative values in relation to real-world contexts. - I can plot negative rational values given a real-world context.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How do I interpret positive and negative rational numbers in real-world contexts? - How can a number line be used to plot positive and negative rational numbers in real-world contexts? - What is the meaning of zero in various real-world contexts?			
 Lesson Narrative	In this lesson, students will interpret positive and negative values in a real-world context, including plotting the values on a number line. Students interpret and plot values related to sea level, finance, weather, and football during the lesson.			
 Component Summary	3.5.1 Warm-Up (~5 min.)	Reflect on biography of a NASA mathematician (<i>SE p. 98</i>)	3.5.4 Collaboration (5-10 min.)	Collaboratively practice determining and plotting values related to temperature (<i>SE p. 101</i>)
	3.5.2 Collaboration (5 min.)	Exploration of elevation with small groups (<i>SE p. 99</i>)	3.5.5 Try It (10 min.)	Practice plotting and interpreting rational numbers in real-world contexts (<i>SE p. 102</i>)
	3.5.3 Guided Instruction (10 min.)	Students learn how to relate positive and negative values to the real-world contexts (<i>SE p. 100</i>)	3.5.6 Your Turn! (5 min.)	Additional practice interpreting rational numbers in real-world contexts (<i>SE p. 104</i>)
	3.5.7 Wrap-Up (~5 min.)	Demonstrate understanding of interpreting and plotting real-world values (<i>SE p. 104</i>)		
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> - Elevation - Deposit - Withdrawal - Expense <u>Additional Terms:</u> N/A		(Optional) Dry erase boards 3.5.1 Warm-Up contains a video to accompany the question prompt. - Prior to question 3 in 3.5.3 Guided Instruction, share pictures related to the Samuel L. Gravely Jr.'s important contributions to the Navy.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may incorrectly say both “negative” and “below” or “positive” and “above” when describing the elevation relative to sea level. - Students may not be able to compare Celsius temperatures to Fahrenheit temperatures or understand that 0°C, not 32°F, is the freezing point. - Students may get stuck with zero representing no difference from the monthly average rainfall in 3.5.5 Try It question 3.	- Facilitate a conversation about the use of the negative symbol in conjunction with words that represent negative values (i.e., below, loss, withdraw) to help students learn how to describe the meaning of values in context correctly. - Explain the Celsius temperature scale to students and show how to represent temperatures on a vertical number line. - Remind students to include the units (i.e., yards) in their answers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Think-Pair-Share	MA.K12.MTR.7.1: Real-World Contexts
	In 3.5.2 and 3.5.4 Collaborations, students initially complete the questions and then discuss their work with a partner. This discussion builds student confidence and understanding.	Throughout this lesson, students use real-world scenarios to interpret and plot positive and negative numbers. Students interpret and plot values related to sea level, finance, weather, and football.
Lesson Notes:		

3.5.1 Warm-Up: Mathematician Katherine Johnson

Component Overview: Students learn about and then reflect on the career of Katherine Johnson, a NASA mathematician.



3.5.1 Warm-Up: Mathematician Katherine Johnson

Born in 1918, Katherine Johnson was always fascinated by numbers. Graduating from high school at 14 and from



college at 18, she went on to become one of the first African American mathematicians with the job title of “computer” at NASA, where she worked for three decades. As a computer, she personally checked the calculations made by electronic computers. Katherine calculated the launch window for America’s first human space flight. Her calculations were critical to the success of the Apollo Moon landing program and the start of NASA’s Space Shuttle program. In 2015, she received the Presidential Medal of Freedom from President Obama. In its official announcement, the White House cited Katherine’s influence on every major space program from Mercury to the Shuttle Program.



Do you know a mathematician, statistician, actuary, or engineer? Consider inviting them to be a guest speaker in your classroom. Additionally, consider having students return to this lesson later in the year for project-based learning around this career profile.

1. What stands out to you the most about Katherine Johnson?

Answers will vary.

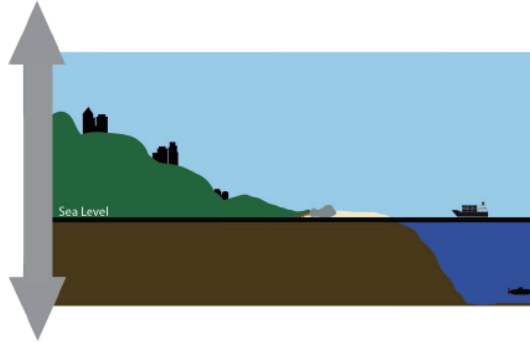
3.5.2 Collaboration: Exploring Elevation

Component Overview: Students work collaboratively as they explore how to determine and plot elevations above and below sea level. Use the image to show students how sea level represents zero and the distance above and below sea level can be represented by positive and negative numbers on a vertical number line.



3.5.2 Collaboration: Exploring Elevation

The picture shows different locations relative to sea level.



- Describe how the picture can be used to explain the word *elevation*.
Answers will vary. The grey arrow on the side shows the elevation of different things in the picture, since elevation is how high or low something is.
- Ask two classmates for their description of the word elevation. Write their description in the table. *You are the only person who should write in the first column of the table.* Have your classmate initial next to your summary of their description stating that your summary is correct.

My Summary of _____'s Description of Elevation	Initials
<i>Answers will vary.</i>	
My Summary of _____'s Description of Elevation	Initials

- Using the picture, name one thing that is below sea level.
Answers will vary. Examples: Dirt/Mud, water, submarine, etc.
- Name one thing that is above sea level.
Answers will vary. Examples: Grass, sky, ship, etc.



For some students, this may be the first time they are hearing or discussing the word elevation. Listen as students collaborate to determine who has prior knowledge of this concept or who develops understanding through their collaboration with peers in this component. The term will be formally defined in the next component, so allow students to focus on sharing what they already know about elevation without overemphasizing that they arrive at a “correct” definition in this component.



- What value do you think sea level represents on the number line?
- What type of values are above sea level on the number line?
- What type of values are below sea level on the number line?

3.5.3 Guided Instruction: Positive and Negative Values in the Real World

Component Overview: Students learn to relate positive and negative rational numbers to the contexts of elevation and finance. Prior to question 2, share pictures related to the Samuel L. Gravely Jr.'s important contributions to the Navy.



3.5.3 Guided Instruction: Positive and Negative Values in the Real World

Elevation is the vertical distance a location or object is from a given level, such as sea level.

Elevation is represented using positive and negative values.

1. If sea level is equivalent to an elevation of 0 units, then what type of value is an elevation that is below sea level?

A negative value.

2. Write each situation as an integer. Plot it on the vertical number line.

- a. A hiker is 37 feet above sea level.

37

- b. A diver is 19 feet below sea level.

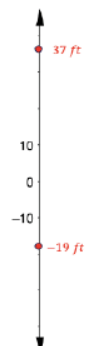
-19

- c. How are these two integers related?

Answers will vary. Both values represent position relative to sea level.

- d. What does 0 represent in this situation?

Sea level.



3. Samuel L. Gravely, Jr. pioneered the way with a plethora of firsts for African Americans in the military. Gravely was the first African American Navy Vice Admiral, first African American to command a Navy warship, first African American to command a warship during combat, first African American to command a Navy fleet, and first African American to obtain Flag Rank in the military. If the Navy fleet included a submarine that was submerged 780 feet below sea level, which students' interpretation of the situation is correct? Circle the name of the student with the correct answer. For any statement that has an error, write a corrected statement.



Consider having students read the vocabulary words aloud to provide an auditory entry point. A visual entry point is provided through the plotting of values on number lines.

3.5.3 Guided Instruction: Positive and Negative Values in the Real World (continued)

Tianna	Keagan
The depth of the submarine is -780 feet below sea level. <i>Answers will vary.</i> <i>The negative in front of 780 indicates that the submarine is below sea level, so it is not correct to say -780 below sea level. It should either be "-780 feet" or "780 feet below sea level," but not both.</i>	780 feet below sea level can be represented by the integer -780.

A bank **deposit** is the amount of money a person gives the bank for safekeeping in their account.

A bank **withdrawal** is the amount of money removed from a bank account.

An **expense** is a cost or the amount of money spent.

4. The following people have money in their bank accounts. Determine the positive and negative values that represent the effect each situation has on their account balance.

Real-World Context	Value
Paul paid \$190.90 in taxes.	$-\$190.90$
Marie made a deposit of \$50.	$\$50$
Tyler made a withdrawal of \$200.	$-\$200$
Hannah gets paid an income of \$500 per week that she deposits in her account.	$\$500$
Kaden pays \$758.91 in monthly expenses.	$-\$758.91$



This is the first unit where students are working with signed values. Use a think aloud to model how to answer question 3 since the students will not necessarily have the prior knowledge to recognize what is wrong with Tianna's response. They will have more practice with this concept over the next few lessons.



Encourage students to create a reference card with vocabulary words and examples. Students can use the reference cards to complete the subsequent question independently.

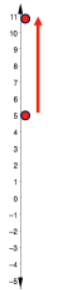
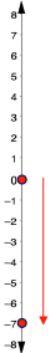
3.5.4 Collaboration: Above or Below Zero

Component Overview: Students collaborate to interpret the meaning of temperature changes using a number line. Students must model that change on the number line in addition to answering a question about the line to check for understanding regarding plotting values.



3.5.4 Collaboration: Above or Below Zero

1. Represent each change in temperature on the number line. Answer the question for each situation.

At noon, the temperature was 5 degrees Celsius. By late afternoon, the temperature had risen by 6 degrees Celsius.		What was the temperature late in the afternoon? 11 degrees Celsius
The temperature was 8 degrees Celsius at midnight. By dawn, the temperature had dropped by 12 degrees Celsius.		What was the temperature at dawn? -4 degrees Celsius
Water freezes at 0 degrees Celsius, but the freezing temperature can be lowered by adding salt to the water. A student discovered that adding half a cup of salt to a gallon of water lowers its freezing temperature by 7 degrees Celsius.		What is the freezing temperature of the gallon of salt water? -7 degrees Celsius



If necessary, explain the Celsius temperature scale to students and show how to represent temperatures on a vertical number line.



Students initially complete the questions and then discuss their work with a partner. This discussion builds student confidence and understanding.

3.5.4 Collaboration: Above or Below Zero (continued)

Numbers below 0 are useful in contexts other than temperature.

2. With your partner, discuss and list other examples where numbers below zero can be used to represent a situation.

Answers will vary. Examples: elevation, money in a bank account.



Expect answers from students such as elevation (“below sea level”) and bank accounts.

3.5.5 Try It: Representing the Real World With Positive and Negative Values

Component Overview: Students practice interpreting and plotting real-world values related to finance, football, and rainfall with a partner. Circulate while students work to support struggling students, monitor student progress, and select which students you will have share during the debrief.



3.5.5 Try It: Representing the Real World with Positive and Negative Values

1. In a bank account, every time money is added or taken out of the account, a new balance is calculated. What does each of the following mean?
 - a. an account balance of $-\$23.56$
Answers will vary.
The account is overdrawn by \$23.56.
 - b. an account balance of $\$143.21$
Answers will vary.
There is \$143.21 available in the account.
 - c. an account balance of $\$0$
Answers will vary.
There is no money in the account.
2. The change in position of the ball during each play of a football game is measured in yards.
 - a. What integer best represents a gain of 5 yards?
+5
 - b. What integer best represents a loss of 15 yards?
-15
 - c. What does the number 0 represent in this context?
Answers will vary.
No loss or gain of yards.
 - d. Plot all three values on the number line.



If students struggle determining if the numeric value is positive or negative, prompt them to identify the key terms. Ask students what clues the terms provide regarding the sign of the number.

3.5.5 Try It: Representing the Real World With Positive and Negative Values (continued)

3. The average monthly rainfall for a city was calculated based on records kept over the past 50 years. The chart shows the difference between the average monthly rainfall and the rainfall amount this year for these months.

Month	Difference From Monthly Average (in inches)
January	$-2\frac{3}{4}$
February	$1\frac{5}{8}$
March	0
April	$-\frac{3}{8}$
May	$4\frac{1}{4}$



Which month had the greatest change from the average monthly rainfall, and how do you know? What does zero represent in the rainfall chart?

Explain what the value in the chart means for the following months in the context of this problem. Then, plot and label each month's value on the number line.

Month	Interpretation	
February	<i>Answers will vary. In February, it rained $1\frac{5}{8}$ inches more than the average rainfall typical for February.</i>	
March	<i>Answers will vary. In March, it rained exactly the typical average amount for March.</i>	
April	<i>Answers will vary. In April, it rained $\frac{3}{8}$ less than the average rainfall typical for April.</i>	

3.5.6 Your Turn!

Component Overview: Students independently practice interpreting rational numbers in real-world contexts. Be sure to remind students to include units in their responses.



3.5.6 Your Turn!

1. Write a value that represents each description.

- | | |
|-----------------------------------|----------------|
| a. an expense of \$40.23 | <u>-40.23</u> |
| b. a deposit of \$225.89 | <u>225.89</u> |
| c. 16,000 feet above sea level | <u>16,000</u> |
| d. a temperature increase of 14°F | <u>14</u> |
| e. a withdrawal of \$271.49 | <u>-271.49</u> |
| f. a gain of 17 yards | <u>17</u> |

2. Read each statement and select the correct description of each situation.

a. A diver is 60 feet below the surface of the ocean.

- ☐ The depth of the diver is 60 feet below the ocean's surface.
- ☐ The diver is -60 feet below the ocean's surface.

b. The elevation of the bottom of a boat with respect to sea level is given as -5 feet.

- ☐ The bottom of the boat is 5 feet above sea level.
- ☐ The bottom of the boat is 5 feet below sea level.



Do your students learn best by co-creating their learning contexts? Consider offering students the opportunity to create their own sentence frames with two options to choose from, then trade with a partner.

3.5.7 Wrap-Up: Ticket Out the Door

Component Overview: In this formative assessment, students demonstrate their understanding of interpreting and plotting rational numbers in real-world contexts.



3.5.7 Wrap-Up: Ticket Out the Door

Keisha observes that one location is 55.2 feet above sea level and another is 46.8 feet below sea level.

1. What rational number represents a location of 55.2 feet above sea level?

55.2 or $55\frac{2}{10}$ or $55\frac{1}{5}$

2. What rational number represents a location of 46.8 feet below sea level?

-46.8 or $-46\frac{8}{10}$ or $-46\frac{4}{5}$

3. Plot and label each rational number on the number line.



4. What does zero represent in this situation?

Sea level

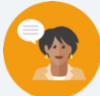


Encourage students to share their reasoning for what sea level represents by depicting their interpretations of the situation. Prompt students to create what each location could actually represent, as context could contribute to better conceptual understanding.

3.6 Comparing a Quantity and Its Opposite in Real-World Context

Lesson Introduction

 Benchmark	MA.6.NSO.1.2 Given a mathematical or real-world context, represent quantities that have opposite direction using rational numbers. Compare them on a number line and explain the meaning of zero within its context.		 Learning Targets	- I can compare a quantity and its opposite on a horizontal or vertical line and explain the meaning of zero within the context. - I can use the structure of a horizontal and vertical number line to plot rational numbers from a real-world context.
 Benchmark Coherence	Building On N/A		Working Toward MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.	
 Essential Questions	- How do you plot opposites on vertical or horizontal number lines? - What can zero represent in various real-world situations? - How does distance from zero compare for a rational number and its opposite?			
 Lesson Narrative	In this lesson, students determine the opposites of rational numbers from a context and represent them on vertical and horizontal number lines. Students will also explain the meaning of zero and opposites within various contexts.			
 Component Summary	3.6.1 Warm-Up (~5 min.)	Write and order integers that represent elevations from images (<i>SE p. 105</i>)	3.6.4 Bring It Together (5 min.)	Describe the relationship between rational numbers and their opposites when plotted on number lines (<i>SE p. 109</i>)
	3.6.2 Collaboration (10 min.)	Identify and plot rational numbers and their opposites (<i>SE p. 106</i>)	3.6.5 Your Turn! (5 min.)	Practice interpreting and plotting rational numbers, their opposites, and the meaning of zero in context (<i>SE p. 109</i>)
	3.6.3 Try It (10-15 min.)	Plot and label rational numbers and their opposites on vertical and horizontal number lines (<i>SE p. 107</i>)	3.6.6 Wrap-Up (~5 min.)	Demonstrate understanding of the lesson learning targets (<i>SE p. 110</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Opposite - Rational number		(Optional) Large paper (Optional) Markers	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions Students may not know the definitions of “withdrawal” and “deposit” or make sense of a deposit as losing money, as students may say, “The money would be leaving your hands so that is a loss.”	Things to Consider Check for understanding of the meaning of zero for each real-world example. For example, in 3.6.3 Try It, be sure students interpret zero as no change in account balance (not a balance of zero).
	Literacy and Language Routines Compare and Connect In 3.6.2 Collaboration and 3.6.4 Bring It Together, students make connections and comparisons among rational numbers, their opposites, and zero. Facilitate discussion about the meaning of zero in each context, as well as the comparisons and connections students make.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Students represent rational number opposites several ways, including on number lines (horizontal and vertical), in tables (using numbers and words), and written descriptions in the 3.6.6 Wrap-Up and 3.6.3 Try It.
	 Differentiate Instruction Consider the following strategies for differentiating instruction in this lesson: In 3.6.4 Bring It Together, students are prompted to provide a written description explaining the relationship between the opposites. Consider having students also create a real-world example of opposites plotted on a number line to demonstrate their understanding.	
Lesson Notes:		

3.6.1 Warm-Up: Ordering Elevations

Component Overview: Students will analyze five pictures to determine the appropriate positive or negative integer that best represents the location. Students will then order the integers from least to greatest.



3.6.1 Warm-Up: Ordering Elevations

Four images from locations across the United States are shown.

- Write an integer that represents each elevation.

a.



Integer: **345**

b.



Integer: **-100**

c.



Integer: **5046**

d.



Integer: **-178**



Consider challenging students by prompting them to research and provide location(s) that would have opposite elevations of the examples in the pictures in questions 1a-1d. Students can also discuss if they can find opposite elevation examples in the same major regions of the U.S.

- Order the letter of the pictures from lowest elevation to highest.

d. , b. , a. , c.

3.6.2 Collaboration: A Quantity and Its Opposite

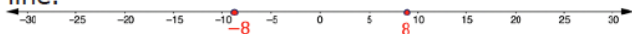
Component Overview: Students collaboratively determine rational numbers, opposites, and the meanings of zero from real-world situations. Students also plot zero and the opposite values on number lines.



3.6.2 Collaboration: A Quantity and Its Opposite

1.

- a. Plot the number 8 and its opposite on the number line.



- b. What do you notice about the distance of 8 and its opposite from zero?

Answers will vary. 8 and its opposite, -8, are the same distance from zero in opposite directions.

2. Complete the table for each situation given.

	78.3 Feet Above Sea Level	32°C Below Zero	Withdrawal of \$27.68
Rational Number	78.3	-32	-27.68
Opposite	-78.3	32	27.68
Meaning of Zero	Sea Level	Freezing Point 0°C	No change in account balance.

3. Omar is at the bookstore on Florida Avenue buying a book about training dogs. He also plans to visit Nolan's Pet Shop and Rover's Pet Shop to compare prices for a new leash for his dog. All three stores are on the same street. Nolan's Pet Shop is 7 blocks west of the bookstore. Rover's Pet Shop is 7 blocks east of the bookstore.

- a. Model the locations of the three stores in relation to each other on the number line.



- b. What does zero represent in this situation?
- c. Which pet store is farther away from the bookstore? Explain how you know.

Zero represents the location of the Bookstore.

Answers will vary. Both pet shops are the same distance from the Bookstore. Both locations are 7 blocks away from the bookstore.



Compare and Connect

Students make connections and comparisons among rational numbers, their opposites, and zero. Facilitate discussion about the meaning of zero in each context as well as the comparisons and connections students make.



- How does the distance from zero compare for a rational number and its opposite?
- What does zero represent in this context?

3.6.3 Try It: A Quantity and Its Opposite

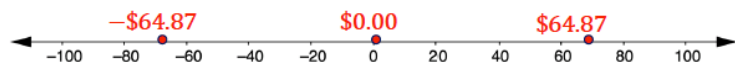
Component Overview: Students plot and label rational numbers and their opposites on vertical and horizontal number lines. Students interpret the meaning of 0 and opposite values in context.



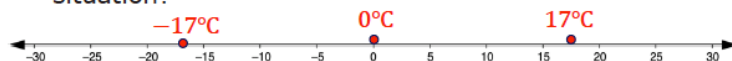
3.6.3 Try It: A Quantity and Its Opposite

- On the number line, plot and label a deposit of \$64.87 and a withdrawal for the same amount from a bank account. What does 0 represent in this situation?

Answers will vary. Zero represents no change in the account balance.



- On the number line, locate and label 17°C below zero and 17°C above zero. What does 0 represent in this situation?



Zero represents a temperature of 0°C , freezing point.

- Carlos is at the ice cream shop. His house is 9.5 blocks north of the shop. The park is 9.5 blocks south of the ice cream shop.
 - Plot and label all three places on the number line.
 - When at the ice cream shop, is Carlos closer to the park or his house?

Answers will vary. The ice cream shop is equal distance to both locations. Ice cream shop represents zero.



Facilitate conversations about the fact that zero represents the change in account balance, not the amount in the account. Conversations should also include the meaning of a deposit and withdrawal.



- Is North up, down, left, or right? What about South?*
- What does zero represent in this situation?*
- How does distance from zero compare for a rational number and its opposite?*

3.6.3 Try It: A Quantity and Its Opposite (continued)

4. Write the rational number that represents the description, the rational number's opposite, and the meaning of the opposite in context of the description.

Description	Rational Number	Opposite	Meaning of Opposite
An atom's positive charge of 7	7	-7	An atom's negative charge of 7
A deposit of \$35.84	35.84	-35.84	A withdrawal of \$35.84
3,540 feet below sea level	-3540	3540	3540 feet above sea level
A rise of 41.6°C	41.6	-41.6	A fall of 41.6°C
A loss of 13.3 pounds	-13.3	13.3	A gain of 13.3 pounds



Consider having students write a story that includes all phrases and values. Creativity will be key in writing a cohesive narrative that incorporates all elements.

5. With a partner, choose one description and describe what 0 could represent in that situation.

Answers will vary.

An atom with no charge.

No change in an account balance.

A location at sea level.

No change in the temperature.

No change in weight.

3.6.4 Bring It Together: A Quantity and Its Opposite

Component Overview: Students work with a partner to describe the relationship between rational numbers and their opposites when plotted on number lines. Select a few groups to present their responses during the whole-class debrief discussion.



3.6.4 Bring It Together: A Quantity and Its Opposite

Work with a partner to write answers to the following questions.

1. What is the relationship between any number and its opposite when plotted on a number line?
Answers will vary. A number and its opposite are the same distance from zero on a number line, but on opposite sides of 0.
2. How can you use this relationship to locate the opposite of a given number on the number line?
Answers will vary. To find the opposite of a given number on a number line, find the same distance, away from zero, in the opposite direction.



Compare and Connect

Students make connections and comparisons among rational numbers, their opposites, and zero. Facilitate discussion about the meaning of zero in each context, as well as the comparisons and connections students make.

3.6.5 Your Turn!

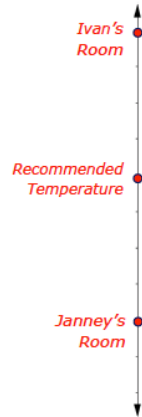
Component Overview: Students practice determining and plotting rational numbers, their opposites, and the meaning of zero in context.



3.6.5 Your Turn!

1. Janney notices that her room's temperature is 4°F below the recommended temperature, and Ivan's bedroom temperature is 4°F above the recommended temperature. Plot the values on the number line and explain what zero represents in this situation.

Answers will vary. Zero represents the recommended temperature.



2. Finn wants to visit his friend Simon who lives 22.7 blocks west. He also wants to visit his grandparents who live 22.7 blocks east.
 - a. Plot and label the locations of Simon and Finn's grandparents on the number line.



- b. What does zero represent in this situation?
Answers will vary. Zero represents Finn's house.
- c. Does Finn live closer to his grandparents or Simon?
Answers will vary. Finn's house is equal distance to both locations.



- *Is the recommended temperature actually zero?*
- *What does that make you think about when engaging with absolute value around numbers that are not zero?*
- *Can a number line still be used?*

3.6.6 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate understanding of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.



3.6.6 Wrap-Up: Ticket Out the Door

Complete the following in the table.

- Write the rational number that represents the situation.
- Write the opposite of the rational number.
- Represent the rational number and its opposite on the number line with appropriate labels.
- Explain the meaning of zero in the situation.

	A Deposit of \$9.34	17°C Below Zero
Rational Number	9.34	-17
Opposite	-9.34	17
Number Line		
Meaning of Zero	There was no change in the account balance.	Freezing point. 0°C



Challenge students to create a context other than what was already used in the lesson and provided here. If students are stuck, prompt students to determine 0 as a certain location in their neighborhood and choosing a new location for the value of 9.34. For example, if their house is 0, then 9.34 could be the distance to the park, while -9.34 could be the distance to their local store (with units).

3.7 Absolute Value

Lesson Introduction

 Benchmark	MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.		 Learning Targets	- I can find the absolute value of a rational number. - I can find the absolute value of a value and interpret it within the context.
 Benchmark Coherence	Building On N/A		Working Toward MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value.	
 Essential Questions	- How can you determine any number's distance from zero on a number line? - What is the relationship between the absolute value of opposites? - What represents zero in the real-world situation? - How do you determine and interpret the absolute value in real-world situations? - How do you determine the absolute value of a number?			
 Lesson Narrative	Students formally define and make meaning of absolute value in this lesson introducing absolute value. Students practice identifying absolute value and plotting it on a number line, then extend their working definitions of absolute value to include sense-making in real-world contexts			
 Component Summary	3.7.1 Warm-Up (~5 min.)	Justify which stack of coins they would rather have (<i>SE p. 111</i>)	3.7.4 Guided Practice (10 min.)	Guided practice determining the absolute value using a number line (<i>SE p. 113</i>)
	3.7.2 Exploration (5-10 min.)	Exploration of distance to zero and opposites using a number line (<i>SE p. 112</i>)	3.7.5 Collaboration (10 min.)	Determine and interpret absolute value and opposites in real-world situations (<i>SE p. 114</i>)
	3.7.3 Guided Instruction (5 min.)	Learn the meaning of absolute value (<i>SE p. 113</i>)	3.7.6 Your Turn! (5-10 min.)	Additional practice determining the absolute value using a number line and interpreting absolute values and opposites in real-world situations (<i>SE p. 116</i>)
	3.7.7 Wrap-Up (~5 min.)	Demonstrate understanding of lesson learning goals (<i>SE p. 116</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Absolute value - Speed <u>Additional Terms:</u> N/A		Number line templates	
			Additional Preparation 3.7.2 Exploration prompts students to use a bug cutout in the number line example.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle estimating the location of fractions and decimals when plotting rational numbers on the number line. - Students may incorrectly think that the distance of a negative integer from zero is a negative number or that the absolute value of a number is its opposite. - Students may believe larger negative numbers also hold larger value (i.e., -7 is greater than -6).	- Avoid defining absolute value as the positive value, as this does not build conceptual understanding, which will have impacts for their understanding of adding and subtracting integers in Unit 6. Focus instead on students making sense of absolute value as the number's distance from zero. - Students may benefit from having number lines to reference throughout the lesson. - If time is an issue, consider assigning 3.7.6 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 3.7.3 Guided Instruction, ask students what they notice and wonder about the absolute values of opposite numbers. Facilitate a discussion to ensure students understand why the absolute values of opposites are equal.	MA.K12.MTR.4.1: Discussion and Reflection 3.7.4 Guided Practice provides an opportunity for students to discuss student work. Students discuss how it could be possible for more than one student answer to be correct to a single problem. Students will build a stronger understanding of the concepts through this discussion.
 Differentiate Instruction	3.7.2 Exploration provides kinesthetic entry points for students who better make sense of concepts using manipulatives and real-world spatial awareness. Consider differentiated approaches to this activity based on your students. - Utilize the bug cutout and number line for students to move along the line. - Construct a life-size number line with students acting as the bugs.	
Lesson Notes:		

3.7.1 Warm-Up: Would You Rather?

Component Overview: Students determine which stack of coins they would prefer. Then students justify their selection.



3.7.1 Warm-Up: Would You Rather

1. Would you rather have Pile A or Pile B? Justify your reasoning.



Pile A	Pile B
14 pennies	2 pennies
7 nickels	4 nickels
2 dimes	2 dimes
1 quarter	2 quarters

Answers will vary. I would rather have Pile A because Pile A has \$0.94 and Pile B has \$0.92.



Contemplate, then Calculate

Use this instructional routine to position students as thinkers first, thinking about their approach before immediately jumping into the calculations. This routine provides space and opportunity for meta-cognition, or thinking about their thinking, which ultimately contributes toward fluency.

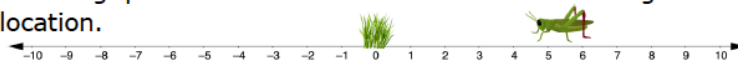
3.7.2 Exploration: Jumping Bugs

Component Overview: Students explore the distance to zero of several integers, including opposites, using a number line. The exploration occurs through the story of a bug that jumps to the left and the right. The bug's home is at zero.



3.7.2 Exploration: Jumping Bugs

A bug is jumping around on the number line. His home is located at 0. Use a cut out of the bug to answer the following questions about the bug's location.



1. The bug starts at 5 and jumps 3 units to the right.
 - a. What is the bug's new location?
8
 - b. How far away from home is the bug?
8 jumps
2.
 - a. If the bug starts at 2 and jumps 5 units to the left, what is the bug's new location?
-3
 - b. How far away from home is the bug?
3 jumps
3. Mr. Rodriguez told his class to begin with the bug at home. Then the bug should jump 4 units. When Micah and Brianna did this, they discovered that their bugs were in two different locations. Discuss with your partner how this may have happened. Write your explanation or draw a picture of what could have happened.

Answers will vary. Micah and Brianna jumped their bugs in opposite directions, landing on -4 and 4.



Try this activity outside to better engage students in making sense of distance between units, direction of units, and more.

3.7.3 Guided Instruction: Absolute Value and Zero

Component Overview: Students complete guided notes as they learn the meaning of absolute value. Consider creating and displaying an anchor chart with instructions, examples, and visual models for students to reference in the subsequent components.



3.7.3 Guided Instruction: Absolute Value and Zero

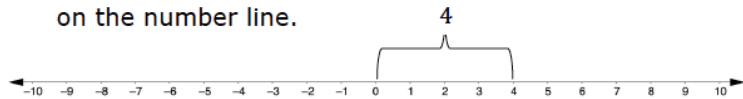
Recall how opposite numbers relate to zero. Complete the explanation below by writing a word in each blank.

1. Pairs of opposite values are the same distance from zero on the number line. They are mirror images of each other, with a line of symmetry through zero.

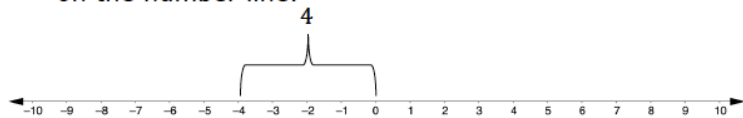


Absolute value is the distance a number is from zero (0) on a number line.

2. The absolute value of 4 is 4 because it is 4 units from 0 on the number line.



3. The absolute value of -4 is 4 because it is 4 units from 0 on the number line.



4. The values 4 and -4 are opposites. Each value is 4 units from zero. A number and its opposite have the same absolute value.



Notice and Wonder

Ask students what they notice and wonder about the absolute values of opposite numbers. Facilitate a discussion to ensure students understand why the absolute values of opposites are equal.

3.7.4 Guided Practice: Absolute Value and Zero

Component Overview: Students practice finding the absolute value of numbers and plotting on a number line.



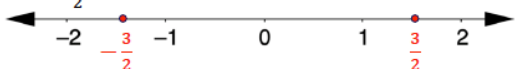
3.7.4 Guided Practice: Absolute Value and Zero

Absolute value is represented using the notation $|a|$. The notation $|a|$ is read, "the absolute value of a ."

1.
 - a. The $|-8| = 8$ because the distance from -8 to 0 is 8 .
 - b. What other number has the same absolute value as -8 ?
 8
 - c. Use the number line to show the other number that has the same absolute value as -8 .



2. On the number line, plot and label all numbers with an absolute value of $\frac{3}{2}$.



3. Hector and his partner, Salih, were given the following two problems to complete.

$$|5.5| = 5.5$$

$$|-3.2| = 3.2$$

3.2 is also correct

Hector and Salih had the same answer for one of the blanks but two different answers for the other. Their teacher told them they both had the correct answers.

- a. Which problem can have more than one correct answer?
The absolute value that is equal to 3.2.
- b. Use the number line to explain how this is possible.
3.2 and -3.2 are the same distance from 0 on the number line, so they both have an absolute value of 3.2.



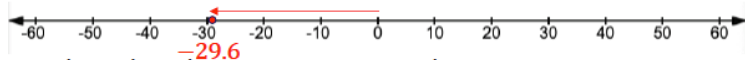
How do you use a number line to determine the distance a number is from zero?

3.7.5 Collaboration: Absolute Value and Opposites in the Real World

Component Overview: Students determine and interpret absolute values in real-world situations. The real-world contexts are related to changes in weight, speed, temperature, and elevation.

3.7.5 Collaboration: Absolute Value and Opposites in the Real World

- A tiger was rescued by The Brevard Zoo in Melbourne, Florida. The zoo veterinarian examined the tiger and determined the animal's weight to be 29.6 pounds below average.
 - What value represents this situation?
 -29.6
 - What is the absolute value of the value that represents the tiger's weight loss?
 $|-29.6| = 29.6$
 - Model the tiger's weight loss on the number line.



- What does 0 represent in this situation?
Answers will vary. Zero represents the change from average weight.
 - If the tiger's weight had been 29.6 pounds above average instead of below, what would be the same about this situation?
Answers will vary. The amount of change would remain the same.
- Simone is doing research on different rides at the Strawberry Festival. She attaches an accelerator to a bumper car to measure the velocity of her car. She recorded the velocities, in miles per hour (mph), in the table.

Speed is the absolute value of velocity.

Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)	Time (seconds)	Velocity (mph)
5	-3.1	20	-2.4	35	-5.2
10	-2.7	25	-6.3	40	3.1
15	4.8	30	2.7	45	-7.8



- When considering the changes in the tiger's weight, what does zero represent?
- How do you model a loss in weight?
- How do you model a gain in average?



Facilitate a conversation about speed and velocity. Provide examples and check for understanding to ensure students understand these terms.

In the context of velocity, the signs represent the direction the bumper car is headed.

3.7.5 Collaboration: Absolute Value and Opposites in the Real World (continued)

- Determine when the bumper car is at the same speed. List the times and the speed of the car at those times.
At 5 seconds and 40 seconds, the car is traveling at 3.1 mph. At 10 seconds and 30 seconds, the car is traveling at 2.7 mph.
 - At what time is the bumper car traveling at the slowest speed?
20 seconds
 - At what time is the bumper car traveling at the fastest speed?
45 seconds
- Before Bryan goes snow skiing, he asks a friend about the weather. The friend says that the temperature is 5°C away from 0. Select all the temperatures that are possible.
 - ☐ -10°C
 - ☒ -5°C
 - ☐ 0°C
 - ☒ 5°C
 - ☐ 10°C
 - A part of the city of New Orleans is 19 feet below sea level. We can use “-19 feet” to describe its elevation, and “|-19| feet” to describe its vertical distance from sea level. In the context of elevation, what do each of the following numbers describe?

Value	Meaning in Context
25 feet	<i>The elevation is 25 feet above sea level.</i>
25 feet	<i>The vertical distance of the elevation from sea level is 25 feet.</i>
-8 feet	<i>The elevation is 8 feet below sea level.</i>
-8 feet	<i>The vertical distance of the elevation from sea level is 8 feet.</i>



Prompt students to reflect on how Bryan’s friend might be more precise in his mathematical language.

- His friend uses the phrase “away from” – why could this have two possible meanings?
- How could this language be more precise to determine the actual temperature?

3.7.6 Your Turn!

Component Overview: Students engage in additional practice determining the absolute value and plotting those values on a number line.

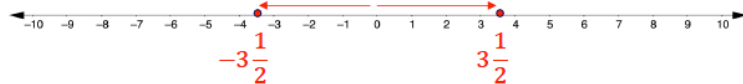
3.7.6 Your Turn!

1. Complete each statement and model it on the number line given.

a. $|-7.8| = 7.8$



b. $|\pm 3\frac{1}{2}| = 3\frac{1}{2}$



2. The absolute value of the elevation of Enterprise, Florida is 19.7 feet. In relation to sea level, write the two values that could represent the elevation of Enterprise, Florida.

19.7 feet or -19.7 feet



Challenge students to write a real-world example to represent the rational numbers and their absolute values from question 1. Remind students to explain what represents zero in their examples. For example, a student's home is zero, school is 3 miles north (+3), the grocery store is 5 miles south (-5).

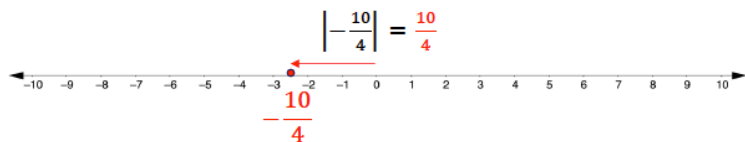
3.7.7 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their mastery of lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.



3.7.7 Wrap-Up: Ticket Out the Door

- Complete the statement and model it on the number line.



- The number -220 represents a withdrawal in dollars from a checking account, and 220 represents a deposit in dollars to the same checking account. What do the absolute values of these numbers mean in terms of this context?

Answers will vary. The absolute value means that the change in the account balance is \$220 in both situations.

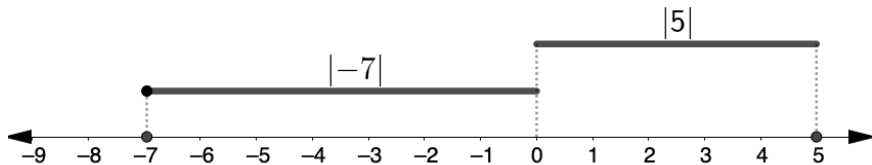


Consider additional information you may want to know from students and add targeted questions to elicit those responses. For example, to elicit responses specific to $\frac{10}{4}$ and $-\frac{10}{4}$ being the same distance from zero, use prompts like: “Write a verbal description for the location of each value on the number line. Compare their location.”

3.8 Comparing and Performing Operations With Absolute Value

Lesson Introduction

 Benchmark	MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute value.		 Learning Targets	- I can compare the absolute value of two numbers. - I can perform operations that involve the absolute values of two numbers.
 Benchmark Coherence	Building On N/A		Working Toward MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.	
 Essential Questions	- How can inequality symbols be used to compare absolute values? - How can number lines be used to compare absolute values? - How do you perform operations with absolute values? - How do you compare absolute values in real-world contexts?			
 Lesson Narrative	Students interpret and compare absolute value in real-world contexts. Lesson activities leverage strategies and tools like number lines and inequality symbols to represent absolute values in context in addition to using the order of operations to evaluate expressions that include absolute values.			
 Component Summary	3.8.1 Warm-Up (~10 min.)	Discuss real-world applications of concepts learned so far in the unit (<i>SE p. 117</i>)	3.8.4 Collaboration (5-10 min.)	Collaboratively compare values and absolute values (<i>SE p. 119</i>)
	3.8.2 Exploration (5-10 min.)	Explore absolute values in context (<i>SE p. 118</i>)	3.8.5 Guided Instruction (10 min.)	Learn how to evaluate expressions that include absolute values (<i>SE p. 120</i>)
	3.8.3 Guided Instruction (10 min.)	Learn how to compare absolute values and consider differences with comparing values (<i>SE p. 118</i>)	3.8.6 Your Turn! (5 min.)	Practice comparing values and absolute values and evaluating expressions that include absolute values (<i>SE p. 121</i>)
	3.8.7 Wrap-Up (~5 min.)	Demonstrate understanding of lesson learning goals (<i>SE p. 122</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Pictures or video clips of real-world contexts	(Optional) Prepare an anchor chart with examples and visual models of key vocabulary and symbols.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle using the correct vocabulary when reading the mathematical statements in 3.8.3 Guided Instruction. - Students may not understand the difference between comparing values and comparing absolute values. - Students may struggle interpreting the meanings of the words elevation, altitude, and sea level.	- Model how to read the mathematical statements in 3.8.3 Guided Instruction. - Circulate and check students' comparisons for accuracy during 3.8.4 Collaboration. - Consider creating and displaying an anchor chart with examples and visual models of key vocabulary and symbols.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns In 3.8.4 Collaboration, students take turns creating mathematical comparison statements and writing the corresponding written statements. Prompt students to explain how they know their statements are true.	MA.K12.MTR.6.1: Reasonableness In 3.8.6 Your Turn!, students compare and evaluate expressions with absolute values. Students consider the reasonableness of their solutions in the given context.
 Differentiate Instruction	Number lines offer spatial and visual strategies for students making sense of absolute value of numbers as their distance from zero. Note the example pictured here from 3.8.5 Guided Instruction. Encourage students to utilize number lines throughout this lesson either to make sense of the numbers or verify their answers.	
		
Lesson Notes:		

3.8.1 Warm-Up: Rational Numbers in the Real World

Component Overview: Students discuss real-world applications of concepts learned so far in the unit.



3.8.1 Warm-Up: Rational Numbers in the Real World

For each of the following problems, ask a classmate for his/her explanation. Once you have received their explanation, write a summary in the space provided in the box. Your classmate should read your summary and initial that it is correct. Repeat this with a different classmate for each of the remaining problems.



Model active listening techniques such as revoicing for students. Remind students to record summaries of their partners' explanations. For example, "So what I am hearing you say is...Is that correct?"

1. Describe the relationship between two numbers that have the same absolute value.

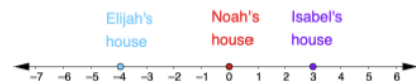
Answers will vary. They are opposites. They have the same distance from zero on a number line.

Initials: _____

2. In an American football game, the offensive team makes gains or losses on each play. Gains are represented by positive values and losses by negative values. What does 0 represent in this context?
- Answers will vary. Zero represents a play where no gains or losses are made.*

Initials: _____

3. Elijah, Noah, and Isabel live on the same street. The locations of their houses are modeled on the number line.



Describe where they live relative to each other in two sentences. Include the words distance and absolute value.

Answers will vary. Elijah's house is a distance of 4 blocks from Noah's house and Isabel's is a distance of 3 blocks. The absolute value of the location of Elijah's house is greater than the absolute value of the location of Isabel's.

Initials: _____

3.8.2 Exploration: Comparing and Interpreting Absolute Value in Context

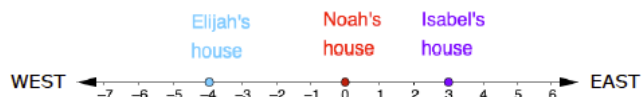
Component Overview: Students interpret and compare absolute values in contexts. Students use a number line to aid in understanding their comparisons. The context explored involves locations east and west of a house.



3.8.2 Exploration: Comparing and Interpreting Absolute Value in Context

- Recall this scenario from the Warm-Up with some additional information.

Elijah, Noah, and Isabella live on the same street that runs west to east. The locations of their houses are modeled on the number line, where each unit represents 1 kilometer.



- Noah's house is located at 0 on the number line model. Represent the distance from Noah's house to both Isabella's house and Elijah's house using absolute value.

	Elijah's House	Isabella's House
Distance from Noah's House	$ -4 $	$ 3 $

- How do the absolute values in the table compare to each other?

$$|-4| > |3|$$

- Isabella lives 3 kilometers to the ☐ east ☐ west of Noah. Elijah lives 4 kilometers to the ☐ east ☐ west of Noah. Isabella lives ☐ closer to ☐ farther from Noah than Elijah does.



During instruction, note the advantages and disadvantages of using horizontal and vertical number lines in different contexts.



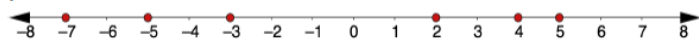
- What does zero represent in the context?
- What represents positive values?
- What represents negative values?

3.8.3 Guided Instruction: Comparing Absolute Values

Component Overview: Students learn how to compare values and absolute values. Students use a number line to evaluate mathematical comparison statements as true or false. Students discuss and write justifications for their answers.

3.8.3 Guided Instruction: Comparing Absolute Values

A number line is shown with the integers $-7, -5, -3, 2, 4,$ and 5 plotted.



1. Discuss with your partner which statements are true. Be prepared to explain your thinking.

Mathematical Statement	Explanation
$-7 < -3$	<i>True. -7 is farther to the left on the number line than -3.</i>
$ -7 $ is 7	<i>True. -7 is 7 units from zero, so the absolute value of -7 is 7.</i>
$ -3 $ is 3	<i>True. -3 is 3 units from zero, so the absolute value of -3 is 3.</i>
$ -7 < -3 $	<i>False. -7 is 7 units from zero while -3 is only 3 units from zero, so -7 is greater than -3.</i>

2. Does the comparison of -7 and -3 match the comparison of $|-7|$ and $|-3|$?
No. $-7 < -3$, but $|-7| > |-3|$.
3. With your partner, discuss why comparing two negative values and comparing the absolute value of the same two negative numbers may be different. Summarize your discussion below.

Answers will vary. Comparing the values of two negative numbers is a comparison of each number's worth, and comparing the absolute values of two negative numbers is a comparison of their distance from zero. For two negative numbers, the number that is farther from zero on the number line has a greater absolute value, but a lower value.



Before students begin the questions, model how to read a few of the mathematical statements. At the end, debrief the key concepts with a discussion to provide an auditory entry point and check for student understanding.

3.8.3 Guided Instruction: Comparing Absolute Values (continued)

4. For each row, use the symbols $>$, $<$, or $=$ to compare each pair of values.

	Inequality	Absolute Value Inequality
Pair 1	$4 > -5$	$ 4 < -5 $
Pair 2	$-3 < 2$	$ -3 > 2 $
Pair 3	$6 > -6$	$ 6 = -6 $

5. What do you notice about the comparisons?

Answers will vary. The comparisons of the values of each pair of numbers use different symbols than the comparisons of their absolute values.

6. When comparing the absolute values of two numbers, numbers that are closer to zero will always have the smaller absolute value.



Consider creating and displaying an anchor chart for students to reference when reading and writing symbols.

Symbol	Meaning in Words
$<$	"is less than"
$>$	"is greater than"
$=$	"is" or "is equal to"
$ x $	"the absolute value of x "

Note that " $<$ " could be interpreted as greater or less than depending on the way students read the inequality statement.

For example: $-2 > -7$ could be read as "negative seven is less than negative two" OR "negative two is greater than negative seven."

3.8.4 Collaboration: Inequality Mix and Match

Component Overview: Students create mathematical statements by selecting two expressions and a comparison symbol from a table. Then, the student's partner represents the mathematical statement with a written mathematical statement. Finally, students switch roles and repeat the activities.



3.8.4 Collaboration: Inequality Mix and Match

Work with your partner to complete the table.

Partner A will select two numbers and one comparison symbol. At least one of the numbers must be written as an absolute value.

Partner B will then write the mathematical statement in words. The following phrases may be helpful: is equal to, the absolute value of, is greater than, is less than. Switch roles until each partner has taken two turns. An example is shown in the first row.

Numbers					Symbol
-0.7	$ -9 $	11	14	$ -8 $	$<$
$-\frac{6}{2}$	-2.2	$ 2.5 $	8	$ 0.7 $	$=$
$ -14 $	$ 20 $	$\frac{7}{2}$	$ 15 $	$-\frac{5}{2}$	$>$

Partner	Mathematical Statement	Statement in Words
Example	$ 3 < 4$	the absolute value of 3 is less than 4
Partner A	Answers will vary.	Answers will vary.
Partner B	Answers will vary.	Answers will vary.
Partner A	Answers will vary.	Answers will vary.
Partner B	Answers will vary.	Answers will vary.



Take Turns

This activity provides students opportunities to take turns creating mathematical comparison statements. To ensure the activity is productive:

- Point students toward the example shown in the workbook.
- Check students' statements for accuracy.
- Facilitate discussions to address misconceptions as needed.



Challenge students who finish early to create real-world examples that can be represented by their mathematical statements.

3.8.5 Guided Instruction: Operations Using Absolute Values

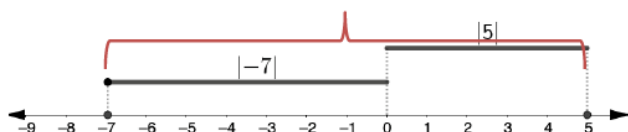
Component Overview: Students learn how to perform operations with absolute values.



3.8.5 Guided Instruction: Operations using Absolute Values

1. Ty'shaun and Gabby are trying to determine the sum of $|-7|$ and $|5|$ and explained their strategies. Ty'shaun modeled his thinking on the number line.

12



Gabby's explanation was "I know $|-7|$ is a distance of 7 from 0 and $|5|$ is a distance of 5 from 0. The sum of 7 and 5 is 12, so the sum of $|-7|$ and $|5|$ is also 12."

Explain which strategy makes more sense to you.

Answers will vary. Gabby's strategy makes sense to me because she just adds the absolute values. Ty'shaun's strategy makes sense to me because he modeled addition on the number line to find the total distance.

2. Find the value of each expression.

$$|-18| - |-13|$$

5

$$12 \times |-4|$$

48

$$\frac{|45|}{|-3|}$$

15

$$|-22| + |31| + |-40|$$

93



Consider modeling how to create and evaluate an absolute value expression. Use the Think Aloud technique to help students understand how you are using the order of operations. Check for understanding before students work independently.

Absolute values are grouping symbols, so they should be evaluated prior to performing other operations. Students would have learned that grouping symbols are evaluated first in elementary school. Avoid referring to PEMDAS or any other acronym. Refer to Unit 13 to preview how the order of operations will be addressed there.



How are Gabby and Ty'shaun's explanations similar?
How are they different?

When do we evaluate absolute values when performing operations?

3.8.6 Your Turn!

Component Overview: Students practice comparing values and absolute values and evaluate absolute value expressions.



3.8.6 Your Turn!

1. Use the symbols $>$, $<$, or $=$ to compare each pair of values.

- a. $|-42| > -25$ d. $|-512| > 284$
 b. $-21 < |18|$ e. $|-176| = |176|$
 c. $|-582| < |-790|$ f. $|-38| > 0$

2. An octopus is 47 feet below sea level at the same time a dolphin is at 108 feet below sea level.

- a. Write a value to describe the position of each sea animal relative to sea level.

Octopus	Dolphin
-47 feet	-108 feet

- b. Use an inequality symbol, $<$ or $>$, to write an inequality comparing the distance each sea animal is from sea level.

$$|-47| < |-108| \text{ OR } |-108| > |-47|$$

- c. What does this inequality statement indicate about the locations of each sea animal?

Answers will vary. The dolphin is farther from the surface of the water than the octopus.

3. Find the value of each expression.

- a. $|-58| + 12 = 70$
 b. $|-13| \times 5 = 65$
 c. $\frac{|-36|}{|9|} = 4$



Prompt students to sketch a number line to represent the situation in question 2 and make meaning in context, if needed.



Students may forget to determine the absolute value before performing the operations in question 3. As you monitor student progress, facilitate a discussion (as needed) about evaluating grouping symbols first.



If students finish early, prompt them to write real-world problems involving comparing absolute values. Ask students to switch problems and compare the absolute values in each other's problems.

3.8.7 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate their mastery of the lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.



3.8.7 Wrap-Up: Ticket Out the Door

- The table shows the elevation, in meters (m), of five locations.

Location	Elevation (m)
Padina Kaundy, Kazakhstan	-132
Laquna del Carbon, Argentina	-105
Zuidplaspolder, Netherlands	-7
Fort Meyers, Florida	2
Britton Hill, Lakewood, Florida	105

For each pair, use $<$, $>$, or $=$ to compare their distances from sea level.

Zuidplaspolder, Netherlands	$ -7 > 2 $
Fort Meyers, Florida	
Britton Hill, Lakewood, Florida	$ 105 = -105 $
Laquna del Carbon, Argentina	
Laquna del Carbon, Argentina	$ -105 < -132 $
Padina Kaundy, Kazakhstan	



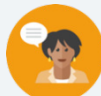
Consider a project-based learning experience for students as homework, extending the 3.8.7 Wrap-Up.

- Map elements of your bedroom, with the level of your pillow or the door handle as “sea level” in the context of elevation.
- Provide the “sea level” of at least five objects in your room.
- Students can use different measuring tools as long as a key is provided.

3.9 Operations With Absolute Value in the Real World

Lesson Introduction

 Benchmark	MA.6.NSO.1.4 Solve mathematical and real-world problems involving absolute value, including the comparison of absolute values.		 Learning Target	I can perform operations that involve the absolute values of two numbers.
 Benchmark Coherence	Building On		Working Toward	
	N/A		M.A.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers, including grouping symbols, whole-number exponents, and absolute value.	
 Essential Questions	- How do you perform operations with absolute values in real-world contexts? - How can number lines be used to create mathematical statements and perform operations with absolute values?			
 Lesson Narrative	Students model various real-world situations with absolute value expressions. Students perform operations with absolute values to solve problems in context.			
 Component Summary	3.9.1 Warm-Up (~5 min.)	Analyze a sale sign for an error (<i>SE p. 123</i>)	3.9.4 Try It (5 min.)	Practice writing and evaluating absolute value expressions in contexts (<i>SE p. 126</i>)
	3.9.2 Collaboration (5-10 min.)	Interpret absolute values in the context of profit (<i>SE p. 123</i>)	3.9.5 Collaboration (10 min.)	Collaboratively write and evaluate absolute value expressions in contexts (<i>SE p. 126</i>)
	3.9.3 Guided Instruction (10-15 min.)	Learn to write and evaluate absolute value expressions to solve problems (<i>SE p. 124</i>)	3.9.6 Your Turn! (10 min.)	Additional practice writing and evaluating absolute value expressions in contexts (<i>SE p. 127</i>)
	3.9.7 Wrap-Up (~5 min.)	Self-assessment of understanding of absolute value learning targets (<i>SE p. 128</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Profit <u>Additional Terms:</u> N/A		(Optional) Chart paper & markers	Consider preparing images or videos of the contexts in the lesson to help students understand the contexts, as needed.

 Teacher Tips	Common Misconceptions - Students may struggle with interpreting the meanings of the words revenue, expenses, and profit. - Students may struggle interpreting the written descriptions of real-world contexts to write absolute value expressions. - Students may add unnecessary (and time-consuming) details to their number lines.	Things to Consider - Consider connecting the terms “gross income,” “expenses,” and “net income” for an individual to the terms “revenue,” “expenses,” and “profit” for a business. - Students have not yet learned to perform operations with integers. Students should determine the absolute values first and then perform the operations.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder In 3.9.1 Warm-Up, students are asked what they notice and wonder about an image of a sale sign. Students should discuss what the error in the sale sign is and why it is not mathematically correct.	MA.K12.MTR.7.1: Real-World Contexts In 3.9.6 Your Turn!, students interpret real-world situations to write absolute value expressions. Consider incorporating elements of students’ personal experiences to enhance their learning and make meaning of absolute value in their own contexts.
 Differentiate Instruction	Consider providing sentence stems or answer choices for questions 4 and 5 in 3.9.5 Collaboration. This will help support students who struggle expressing themselves in writing to demonstrate their understanding.	
Lesson Notes:		

3.9.1 Warm-Up: Math Fail

Component Overview: Students analyze a sale sign for errors. Students explain what the error is and why it is a mathematical error.



3.9.1 Warm-Up: Math Fail

1. What do you notice about the sign?



Answers will vary. The sale price is greater than the original price, so that's not a good sale. Also, the math is not correct because $69.98 - 49 = 20.98$.



Notice and Wonder

Ask students what they notice and wonder about an image of a sale sign. Students should discuss what the error in the sale sign is and why it is not mathematically correct.



Consider challenging students to find more Math Fails as homework or extra credit as they review for the end of the unit.

3.9.2 Collaboration: Absolute Value and Profit

Component Overview: Students work collaboratively to interpret and compare absolute values in context. The context explored involves revenue, expenses, and profit of a lawn mowing business.



3.9.2 Collaboration: Absolute Value and Profit

- Landon starts a lawn-mowing business to earn money. All of his profits are being saved for college.

Profit is the difference between the amount earned (revenue) and the amount spent.

His father lends him a lawnmower to use for free, but he must pay for the gas and other supplies. For each lawn, his expenses are \$7.30, and he earns an average of \$40.00.

- In this situation, revenue is represented with a

☐ positive
☐ negative

value and an expense is represented

by a

☐ positive
☐ negative

value.

- Write a value to represent each situation.

Landon's Finances	Value
Expenses per lawn	-\$7.30
Revenue per lawn	\$40.00

- Use an inequality symbol, $<$ or $>$, to compare the absolute values of Landon's expenses to revenue for each lawn he mows.

$$|\$40.00| > |-\$7.30|$$

- How does the inequality show that Landon will make a profit?

Answers will vary. The revenue he brings in is greater than his expenses, so he will make a profit.

- To make a profit, total expenses of a business must

be ☐ greater than
☐ less than

total revenue. A business is not

profitable when the total

☐ expenses are
☐ revenue is

greater

than the ☐ expenses.
☐ revenue.



Consider having students create reference cards with vocabulary terms from the contexts explored in this lesson. Include definitions, examples, and visual models to improve student comprehension.

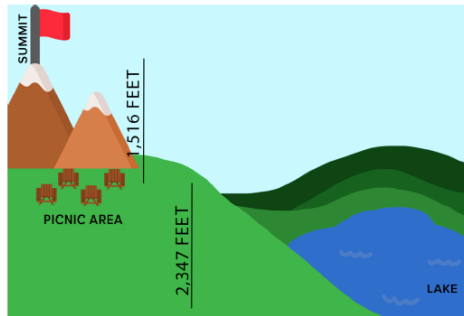
3.9.3 Guided Instruction: Operations With Absolute Value in the Real World

Component Overview: Students learn to write and evaluate absolute value expressions in contexts. The contexts explored include elevation and finance.



3.9.3 Guided Instruction: Operations with Absolute Value in the Real World

- Javier and his family drove to a picnic area on a mountain. From the picnic area the family hiked to the mountain's summit, which was an elevation change of 1,516 feet (ft). They hiked back to the picnic area to eat lunch. After lunch, they hiked down to the lake, which was an elevation change of 2,347 ft.

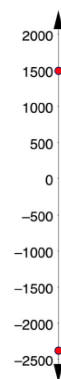


- In the table, write the value that represents the position of each location if the value of the position of the picnic area is 0. Then, plot each value at the estimated location on the number line.

Location	Value
Picnic area	0
Mountain summit	1516
Lake	-2347

- Represent the distance of each location from the picnic area.

Location	Distance from Picnic Area
Mountain summit	1516 ft.
Lake	2347 ft.



- When calculating the distance between elevations, how do you know whether to add the absolute values or subtract them?
- How can the absolute values of a company's expenses and revenues be used to determine whether the company makes a profit?

3.9.3 Guided Instruction: Operations With Absolute Value in the Real World (continued)

- c. What operation would you use with the representations in the table to find the distance from the lake to the mountain summit? *Addition*
 - d. What is the distance from the lake to the mountain summit? *3863 feet*
 - e. What operation would you use to determine how much farther a hike to the lake is than a hike to the mountain summit? *Subtraction*
 - f. How much farther is a hike to the lake than a hike to the mountain summit? *831 feet*
2. Chloe is starting a painting business. The supplies she purchased for the business cost \$825. She will earn a fee of \$80 for each room that she paints. Chloe wants to determine how many rooms she will need to paint before she starts making a profit.

- a.

☐ Fees Chloe earns
☐ Supplies Chloe buys

 increase her profit.
- b.

☐ Fees Chloe earns
☐ Supplies Chloe buys

 decrease her profit.
- c. Write a value in the blank for each absolute value and a word in the remaining blank to complete each sentence.

To find the revenue, Chloe multiplies the number of rooms she paints by $|80|$.

To find the profit, Chloe subtracts $|-825|$ from the revenue.

- d. In the table, complete the mathematical statement and then perform the operations to determine if the number of rooms painted would result in a profit.

Number of Rooms	Mathematical Statement	Profit (Yes or No)
6	$6 80 - -825 $	<i>No</i>
10	$10 80 - -825 $	<i>No</i>
13	$13 80 - -825 $	<i>Yes</i>
16	$16 80 - -825 $	<i>Yes</i>



Consider extending question 2 by asking early finishers the following:

- Find the number of rooms Chloe needs to paint to break even.
- How would the mathematical statement change if Chloe earns a fee of \$85 dollars per room?

3.9.4 Try It: Operations With Absolute Value in the Real World

Component Overview: Students practice writing and evaluating absolute value expressions in contexts. The context explored is locations of students' homes relative to their school. Students use a number line to aid them in answering the questions.



3.9.4 Try It: Operations with Absolute Value in the Real World

Logan, Mia, and Xavier all live on the same street as their school, which runs from east to west.

- Logan lives 4 blocks to the west of the school.
 - Mia lives 5 blocks to the east of the school.
 - Xavier lives 2 blocks to the west of the school.
1. Use the number line to model the position of each house and the school along the street. Label the location of each person's house and the school.



2. Work with your partner to write the mathematical sentence using absolute values. Then, use the mathematical sentence to determine each answer.
 - a. How far does Logan live from Mia?
 $| -4 | + | 5 | = 9 \text{ blocks}$
 - b. How far does Mia live from Xavier?
 $| 5 | + | -2 | = 7 \text{ blocks}$
 - c. How far does Xavier live from Logan?
 $| -4 | - | -2 | = 2 \text{ blocks}$
 - d. Was the same operation used to find each distance?
No



Sample questions to consider to check for understanding as students work:

- How did you determine where to plot the students' homes on the number line?
- How did you determine how far each student lives from school?
- How did you determine the distance between students' homes?
- How did you know whether to add or subtract the absolute values?

3.9.5 Collaboration: Halfway Point

Component Overview: Students work with a partner to determine the appropriate intervals for a number line used to model the elevations of Mount Ojos and the Atacama Trench. Students discuss how to determine the halfway point between the two locations and whether it is above or below sea level.



3.9.5 Collaboration: Halfway Point

Mount Ojos del Salado is the highest mountain in Chile. Its peak is 6,893 meters above sea level. The lowest point of the Atacama Trench, just off the coast of Peru and Chile, is 8,065 meters below sea level.

Work with your partner to answer each question.

1. Determine how to label the number line so that you can approximate a location on the number line that represents Mount Ojos del Salado, Atacama Trench, and sea level.

Answers will vary. Answer shown on number line.

2. Label the number line and place a point at the approximate location for Mount Ojos del Salado, Atacama Trench, and sea level.

Answer shown on number line.

3. Write a mathematical sentence using absolute value to find the difference in elevations between Mount Ojos del Salado and the Atacama Trench.

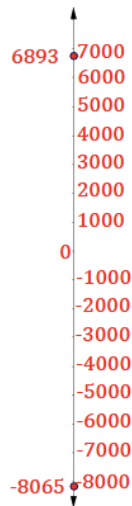
$$|-8065| + |6893| = 14,958$$

4. With your partner, discuss how the halfway point between the peak of Mount Ojos del Salado and the Atacama Trench can be found. Write down all of the strategies that were discussed.

Answers will vary. Divide the total distance by 2, and then add that number to -8,065 or subtract that number from 6,893.

5. Explain how to determine if the halfway point's elevation is above sea level or below sea level.

Answers will vary. The halfway point will be at an elevation below sea level because Atacama Trench is farther below sea level than Mount Ojos del Salado is above sea level.



Consider showing video clips of Mount Ojos del Salado and Atacama Trench or mapping the location of Chile and Peru on a map to increase knowledge in context.



The following questions could facilitate connections for students struggling to discuss or explain what they see:

- What's the total distance between the two elevations?
- What is half that distance? How far is each location from sea level?
- How can you use this information to determine whether the halfway point between the two locations is above or below sea level without performing the calculations?

3.9.6 Your Turn!

Component Overview: Students practice labeling a number line with the elevations of the captain and his fishing net. Students create an absolute value expression to determine the length of the rope. Next, students write and evaluate absolute value expressions to represent debt and temperature changes of chicken in a freezer.



3.9.6 Your Turn!

1. The captain of a fishing vessel is standing on the deck at 23 feet above sea level. He holds a rope tied to a fishing net below him that is underwater at a depth of 38 feet.

- a. Draw a model of this situation using the number line.

Answer shown on number line.

- b. Write a mathematical sentence using absolute value to determine the length of the rope.

$$|23| + |-38| = 61$$



2. Aryan owes \$325.46 for recent purchases he made using his credit card and he owes his parents another \$187.27.

- a. Write a mathematical statement using absolute value that can be used to determine Aryan's total debt.

$$|-325.46| + |-187.27| = 512.73$$

- b. What is the value of Aryan's total debt?

$$\$512.73$$

3. When Brenda places a package of chicken in the freezer, the chicken is at a temperature of 41° Fahrenheit ($^{\circ}\text{F}$). While in the freezer, the temperature of the chicken changes -2°F every hour until it reaches the temperature of the freezer. What is the temperature of the chicken 3 hours after Brenda puts it in the freezer?

$$|41| - 3|-2| = 35^{\circ}\text{F}$$



Considerations to differentiate based on student need:

- Facilitate small group conversations about what key points need to be plotted on the number line.
- Review the number line from 3.9.3 Guided Instruction.
- Prompt students to create an open number line with only the locations of 0, 23, and -38 . Make sure students plot -38 farther from 0 than 23.



- How is a fishing net used?
- How is a credit card used?
- What happens to the temperature of meat when you place it in the freezer?

3.9.7 Wrap-Up: Reflection

Component Overview: Students reflect on their mastery of the unit learning targets related to absolute value. Use the data from this formative assessment to determine which students need remediation before the next lesson.



3.9.7 Wrap-Up: Reflection

1. Check all of the statements below that describe your understanding and ability of working with absolute values.

Answers will vary.

- ☐ I can compare the absolute value of two numbers. I need support on how to compare the absolute value of two numbers.
- ☐ I can perform operations that involve the absolute value of two numbers.
- ☐ I need support to perform operations that involve absolute value.
- ☐ I can interpret comparisons of absolute values in real-world contexts.
- ☐ I need support interpreting comparisons of absolute values in real-world contexts.
- ☐ I can perform operations that involve absolute value when solving real-world problems.
- ☐ I need support to perform operations that involve absolute value when solving real-world problems.



Consider which lessons to review based on students' responses:

- "I can compare the absolute value of two numbers." (Lesson 8)
- "I can perform operations that involve the absolute value of two numbers." (Lesson 8)
- "I can interpret comparisons of absolute values in real-world contexts." (Lesson 7)
- "I can perform operations that involve absolute value when solving real-world problems." (Lesson 9)